Large-Scale Dataset Analysis: 250K Job Records

📑Project Overview

Large-Scale Dataset Analysis: 250K Job Records

We analyzed 250,000 jobs in the fields of: **'AI Engineer', 'Data Analyst', 'Frontend Developer', 'Business Analyst', 'Product Manager', 'Backend Developer', 'Machine Learning Engineer', 'DevOps Engineer', 'Software Engineer', 'Cybersecurity Analyst', 'Data Scientist', 'Cloud Engineer'**

In this project, we analyzed data downloaded from **Kaggle**. We started the project by bringing the data using **pandas** to start working on it.

Part 1: Data cleaning, memory optimization and validation

In this part, we looked for potential issues, such as duplicate strings in different formats. We also monitored the logic of the numbers in the data and optimized it by converting duplicate strings to**a categorical** data type.

We added an important part under the title: **"Beyond the Numbers: How Realistic is this Salary Data"** and we examined whether this data is real or at least realistic.

Finally, in this part, we added a title **"Data Validation"** to define the analysis methodology in the project. We said: In this analysis, we deal with data as purely numerical values within this specific set of data. It is important to note that direct comparisons between countries may be misleading due to the potential for variations in data distribution.

Part 2: Analysis

We performed four main parts of the analysis.

- Correlation Analysis: Years of Experience vs. Annual Salary
- Analyzing the Correlation of Education Level with Salary
- Country-wise Comparison
- Job title comparison

Part 3: Is the data synthetic?

In the third and final part of the project under the title **"Is the data synthetic?"**, we went inside the data and searched if this data is realistic or not, such as: is the data distribution logical and are the numbers inside the data logical.

Dependencies & Data Import

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import seaborn as sns
import numpy as np
```

In [2]:
```
df = pd.read_csv('../Dataset/original_dataset.csv')
```

This dataset is under Creative Commons CC0 (Public Domain Dedication).

I. Data cleaning, memory optimization and validation

In [3]:
```
df.head(10)
```

Out[3]:

	job_title	experience_years	education_level	skills_count	industry	company_
0	AI Engineer	10	Bachelor	2	Healthcare	Med
1	Data Analyst	5	Bachelor	17	Telecom	Sı
2	Frontend Developer	18	PhD	4	Media	Med
3	Business Analyst	19	PhD	13	Retail	Med
4	Product Manager	15	Bachelor	7	Manufacturing	La
5	AI Engineer	0	High School	4	Education	Enterp
6	Backend Developer	6	High School	16	Manufacturing	Enterp
7	Machine Learning Engineer	4	Bachelor	18	Healthcare	Enterp
8	AI Engineer	5	Diploma	14	Telecom	Enterp
9	Frontend Developer	18	PhD	2	Retail	Enterp

In the previous cell, we used `df.head(10)` to display the first 10 rows of the DataFrame. This allows us to quickly inspect the structure of the data.

I-1. 📊 Data Structure & Technical Audit

In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   job_title         250000 non-null  str
 1   experience_years  250000 non-null  int64
 2   education_level   250000 non-null  str
 3   skills_count      250000 non-null  int64
 4   industry          250000 non-null  str
 5   company_size      250000 non-null  str
 6   location          250000 non-null  str
 7   remote_work       250000 non-null  str
 8   certifications    250000 non-null  int64
 9   salary            250000 non-null  int64
dtypes: int64(4), str(6)
memory usage: 30.7 MB
```

A technical inspection of the 250,000 records using df.info() reveals the following:

- Data Integrity
  The dataset is fully complete with no missing values, meaning no imputation is required at this stage.

- Feature Types
  The dataset includes 4 numerical columns (e.g., salary, experience_years) correctly stored as int64, and 6 categorical columns currently represented as strings.

- Memory Optimization
  The dataset currently uses approximately 30.7 MB of memory.
  To improve efficiency, high-frequency string columns such as industry, education_level, and remote_work will be converted to the category data type, reducing memory usage.

I-2. 🔍 Categorical Values Validation (Data Integrity Check)

In [5]:
```
text_cols = df.select_dtypes(include=['object', 'string']).columns

for col in text_cols:
    print(f"--- Unique values in {col} ---")
    print(df[col].unique())
    print("\n")
```

```
--- Unique values in job_title ---
<ArrowStringArray>
[              'AI Engineer',              'Data Analyst',
        'Frontend Developer',          'Business Analyst',
           'Product Manager',         'Backend Developer',
 'Machine Learning Engineer',           'DevOps Engineer',
         'Software Engineer',     'Cybersecurity Analyst',
            'Data Scientist',            'Cloud Engineer']
Length: 12, dtype: str


--- Unique values in education_level ---
<ArrowStringArray>
['Bachelor', 'PhD', 'High School', 'Diploma', 'Master']
Length: 5, dtype: str


--- Unique values in industry ---
<ArrowStringArray>
[   'Healthcare',       'Telecom',         'Media',        'Retail',
 'Manufacturing',     'Education',       'Finance',    'Technology',
    'Consulting',    'Government']
Length: 10, dtype: str


--- Unique values in company_size ---
<ArrowStringArray>
['Medium', 'Small', 'Large', 'Enterprise', 'Startup']
Length: 5, dtype: str


--- Unique values in location ---
<ArrowStringArray>
[      'India',   'Australia',   'Singapore',      'Canada',      'Swede
n',
         'USA', 'Netherlands',      'Remote',     'Germany',          'U
K']
Length: 10, dtype: str


--- Unique values in remote_work ---
<ArrowStringArray>
['Hybrid', 'No', 'Yes']
Length: 3, dtype: str
```

After performing a deep dive into the unique values of each categorical column, we have verified the following:

1. Data Consistency & Cleaning Quality

   No Redundancy: There are no duplicate categories written in different styles (e.g., "Remote" and "remote" are not present; only one consistent format exists).

   No Whitespace Issues: All string values are "trimmed" and clean; there are no leading or trailing spaces

Status: ✅ Production-Ready

Conclusion: Everything is perfectly aligned. The categories are logical, standardized, and clean, providing a robust foundation for the next stage of statistical analysis and visualization.

I-3. 📈 Statistical Summary & Sanity Check

In [6]:
```
df.describe()
```

Out[6]:

	experience_years	skills_count	certifications	salary
count	250000.000000	250000.000000	250000.000000	250000.000000
mean	10.005408	9.997812	2.491928	145718.080524
std	6.060602	5.479288	1.706475	37407.952729
min	0.000000	1.000000	0.000000	31867.000000
25%	5.000000	5.000000	1.000000	119358.000000
50%	10.000000	10.000000	2.000000	143453.000000
75%	15.000000	15.000000	4.000000	169492.000000
max	20.000000	19.000000	5.000000	333046.000000

Through the analysis of Descriptive Statistics for numerical features:

- Range: Salaries range from $31,867 (Minimum) to $333,046 (Maximum).
- There are no "impossible" values (such as negative salaries, 1000 years of experience, or millions of certificates).

Status ⚠️ : There is still an underlying issue that we will address in the section: I-7. 🤔Beyond the Numbers: How Realistic is this Salary Data

I-4. 💾 Initial Memory Consumption (Pre-Optimization)

In [7]:
```
print(df.memory_usage(deep=True))
```

```
Index                  132
job_title            6061551
experience_years     2000000
education_level      3750430
skills_count         2000000
industry             4174893
company_size         3648620
location             3600551
remote_work          2916807
certifications       2000000
salary               2000000
dtype: int64
```

Before applying any data type transitions, we inspected the memory footprint of the

DataFrame using df.memory_usage(deep=True).

1. Analysis of Current State

   Heavyweight Columns: The text-based columns (stored as object) are consuming the most memory. For example:

   ```
   job_title: ~6.06 MB

   industry: ~4.17 MB

   education_level: ~3.75 MB
   ```

2. Technical Observation

   Status: ⚠️ Pre-Optimization Phase

   Issue: Storing repetitive strings as "Objects" is inefficient. Each entry is being stored as a full string pointer, which scales poorly as the dataset grows.

   Observation: The total memory usage is dominated by these categorical strings, which are prime candidates for optimization.

3. Transition Goal

This baseline measurement serves as our Control Group. In the next step, we will convert these 6 string columns to the category type to observe the drastic reduction in memory overhead.

I-5. ⚙️ Data Type Transformation (Memory Optimization)

In [8]:
```
df['job_title'] = df['job_title'].astype('category')
df['education_level'] = df['education_level'].astype('category')
df['industry'] = df['industry'].astype('category')
df['company_size'] = df['company_size'].astype('category')
df['location'] = df['location'].astype('category')
df['remote_work'] = df['remote_work'].astype('category')
```

In this step, we executed a strategic conversion of categorical features to enhance performance and reduce memory overhead.

1. The Engineering Action

We identified 6 columns originally stored as generic Strings (Objects) and explicitly cast them to the category data type:

```
job_title, education_level, industry, company_size,
location, and remote_work.
```

2. Why Category over String?

   Storage Efficiency: Instead of storing the full text for each of the 250,000 rows, Pandas now stores a unique integer key for each category, mapping it to the

original string in a small lookup table.

Computational Speed: Operations such as sorting, grouping, and unique value counting are now significantly faster as they operate on integers "under the hood."

In [9]:
```
print(df.memory_usage(deep=True))
```

```
Index                   132
job_title            250293
experience_years    2000000
education_level      250076
skills_count        2000000
industry             250169
company_size         250074
location             250146
remote_work          250036
certifications      2000000
salary              2000000
dtype: int64
```

Status: ✅ Optimized

```
Result: The data remains logically identical, but the
internal representation is now optimized for large-scale
data analysis and machine learning workflows.
```

In [10]:
```
## df.to_parquet('jobs_data_optimized.parquet')
```

After optimizing the data types and verifying the integrity of the 250,000 records, we exported the final DataFrame to a Parquet file format.

In [11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   job_title         250000 non-null  category
 1   experience_years  250000 non-null  int64
 2   education_level   250000 non-null  category
 3   skills_count      250000 non-null  int64
 4   industry          250000 non-null  category
 5   company_size      250000 non-null  category
 6   location          250000 non-null  category
 7   remote_work       250000 non-null  category
 8   certifications    250000 non-null  int64
 9   salary            250000 non-null  int64
dtypes: category(6), int64(4)
memory usage: 9.1 MB
```

I-6. 📉 Optimization Results

After converting the string-based columns to the category data type, we re-inspected the DataFrame structure to quantify the performance gains.

1. Key Observations

   Data Type Update: All 6 target columns (job_title, education_level, etc.) are now successfully recognized as category instead of object.

2. Memory Usage Impact

   Initial Memory: ~30.7 MB

   Optimized Memory: 9.1 MB

   Efficiency Gain: We achieved a ~70% reduction in memory footprint.

3. Why this matters for Scalability

By reducing the memory usage to less than a third of its original size, we have:

```
Faster Computations: Operations like df.groupby() or
df.pivot_table() will now run much faster because they
process integer codes instead of raw strings.
```

I-7. 🤔Beyond the Numbers: How Realistic is this Salary Data

In [12]:
```
office_based_df = df[df['remote_work'] == 'No']

job_location_median = office_based_df.pivot_table(
    index='job_title',
    columns='location',
    values='salary',
    aggfunc='median'
).fillna(0)

print("Median Salaries by Job Title and Location (Office-Based Only):")
display(job_location_median)
```

```
Median Salaries by Job Title and Location (Office-Based Only):
```

location job_title	Australia	Canada	Germany	India	Netherlands	Remote	Singapor
AI Engineer	160215.0	192125.0	181225.0	112848.0	164525.0	162693.0	158868.(
Backend Developer	130423.5	157828.0	141210.0	90420.5	129058.0	129841.0	128358.(
Business Analyst	114531.5	134063.5	127232.0	79181.0	114380.0	112881.0	113750.(
Cloud Engineer	141148.0	169855.0	155653.0	98713.0	142053.0	142386.0	144156.!
Cybersecurity Analyst	140484.0	166226.0	151249.0	95770.0	137450.5	138998.5	138872.(
Data Analyst	110535.0	133458.0	122220.0	78205.5	112271.0	111394.0	110897.(
Data Scientist	136338.5	164743.0	150207.0	95579.0	136546.0	137947.0	137920.(
DevOps Engineer	140830.0	167385.0	155249.0	98638.0	140120.5	141713.0	139578.!
Frontend Developer	123114.0	148758.0	135843.5	87090.0	122473.0	124232.0	125362.!
Machine Learning Engineer	154571.0	181855.0	168476.5	106114.0	150847.5	152489.5	151614.(
Product Manager	146159.0	176583.5	161170.0	102980.0	147222.0	147296.0	147203.(
Software Engineer	128681.0	156109.5	146192.0	93570.0	130121.0	130353.0	130999.(

This table illustrates the Median Salary per country and job title for office-based roles **(Remote Work = No)**.

Let's take India as a sample from this table. The median salary shown for India does not seem to reflect reality; however, this could be due to a small sample size, which can often be misleading. Let's verify:

In [13]:
```
office_based_df = df[df['remote_work'] == 'No']

job_location_count = office_based_df.pivot_table(
    index='job_title',
    columns='location',
    values='salary',
    aggfunc='count'
).fillna(0)

print("Sample Size (Count) per Job and Location (Office-Based Only):")
display(job_location_count)
```

```
Sample Size (Count) per Job and Location (Office-Based Only):
```

location job_title	Australia	Canada	Germany	India	Netherlands	Remote	Singapore	Swe
AI Engineer	672	715	662	689	689	672	710	
Backend Developer	704	677	745	712	698	680	671	
Business Analyst	720	692	731	691	663	632	667	
Cloud Engineer	670	707	705	667	675	642	714	
Cybersecurity Analyst	735	697	729	753	684	702	693	
Data Analyst	677	705	697	744	693	713	701	
Data Scientist	728	711	685	659	692	711	688	
DevOps Engineer	717	711	688	691	712	692	718	
Frontend Developer	711	691	684	707	709	671	696	
Machine Learning Engineer	700	742	668	715	650	708	700	
Product Manager	772	718	696	697	661	703	736	
Software Engineer	671	702	649	645	687	708	666	

This table displays the sample size (count) of salary records per country and job title for office-based roles (Remote Work = No).

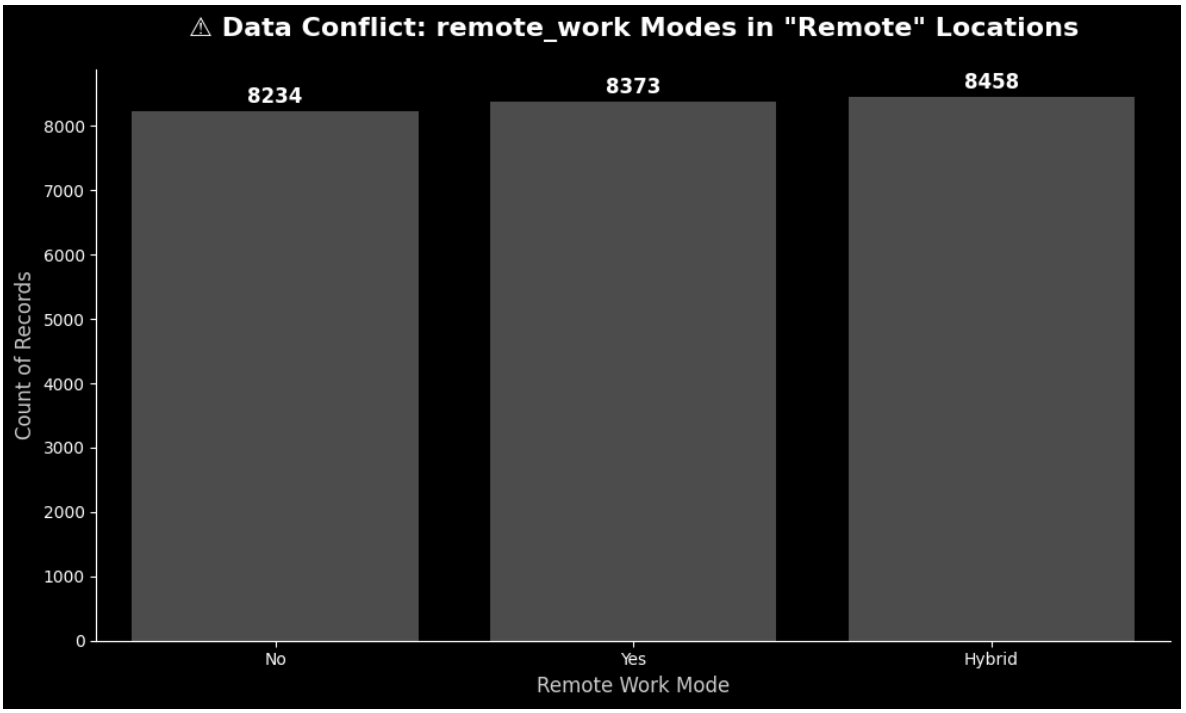
Upon observation, we have a sufficient sample size for job roles in India. Therefore, the unrealistic median salary is not because to a lack of data points. Instead, this suggests two possibilities:

```
- Selection Bias: The dataset may have been sampled from
specific high-paying tiers

- Synthetic Nature: The data might be synthetic
(artificial) and not reflective of real-world labor market
conditions.
```

⚠️ Critical Observation: Logical Data Conflict

Even after filtering the dataset for office-based roles **remote_work = No**, a significant number of records still list "Remote" as their physical location. This represents a logical inconsistency in the data. Which further confirms that this dataset may be synthetic

In [14]:
```
plt.style.use('dark_background')
plt.figure(figsize=(10, 6))

remote_location_df = df[df['location'] == 'Remote']
remote_work_counts = remote_location_df['remote_work'].value_counts().rei

ax = sns.barplot(x=remote_work_counts.index, y=remote_work_counts.values,

plt.title('⚠️ Data Conflict: remote_work Modes in "Remote" Locations',
          fontsize=16, fontweight='bold', color='#ffffff', pad=20)
plt.xlabel('Remote Work Mode', fontsize=12, color='#cccccc')
plt.ylabel('Count of Records', fontsize=12, color='#cccccc')

sns.despine()

for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='baseline',
                fontsize=12, fontweight='bold', color='white', xytext=(0,
                textcoords='offset points')

plt.tight_layout()
plt.show()
```



In [15]:
```
%%html
<div style="padding: 20px;
```

```
            background-color: #1a1a1a;
            border-left: 6px solid #ff4d4d;
            border-radius: 8px;
            color: #ffffff;
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

    <h3 style="color: #ff4d4d; margin-top: 0; display: flex; align-items:
        <span style="margin-right: 10px;">⚠️</span> Critical Alert
    </h3>

    <p style="font-size: 1.1em; line-height: 1.6; margin-bottom: 10px;">
        As this graph illustrates, there is a clear <b>inconsistency</b>:
        <span style="color: #00ffcc; font-family: monospace;">"Remote"</s
        simultaneously categorized as <span style="color: #ffcc00; font-f
        for remote work.
    </p>

    <p style="font-size: 1.1em; line-height: 1.6; margin-bottom: 10px;">
        This confirms a <b>logical discrepancy</b> in the data, strongly
        <span style="text-decoration: underline; color: #ff4d4d;">synthet
    </p>

    <hr style="border: 0; border-top: 1px solid #444; margin: 15px 0;">


</div>
```

⚠️ Critical Alert

As this graph illustrates, there is a clear **inconsistency**: many records labeled as "Remote" in the location column are simultaneously categorized as "No" for remote work.

This confirms a **logical discrepancy** in the data, strongly suggesting that the dataset is synthetic or artificially generated.

I-8. 📑Data Validation

In [16]:

```
%%html
<div style="padding: 15px;
            background-color: #f8f9fa;
            border-left: 5px solid #2ecc71;
            border-radius: 5px;
            color: #2c3e50;
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

    <h4 style="color: #27ae60; margin-top: 0;">📘 Methodological Note: Nu

    <p style="font-size: 1em; line-height: 1.5;">
        In this analysis, we treat the data as <b>pure numerical values</
    </p>

    <p style="font-size: 0.95em; line-height: 1.5; color: #34495e;">
```

```
        <i>Example:</i> We might have a country with a high average salar

        This might not apply to these data points, but it's just an examp
    </p>

    <hr style="border: 0; border-top: 1px solid #ddd; margin: 10px 0;">

</div>
```

📘 Methodological Note: Numerical Scope & Potential Bias

In this analysis, we treat the data as **pure numerical values** within the scope of this specific dataset. It is crucial to acknowledge that direct cross-country comparisons may be misleading due to **the presence of variables**.

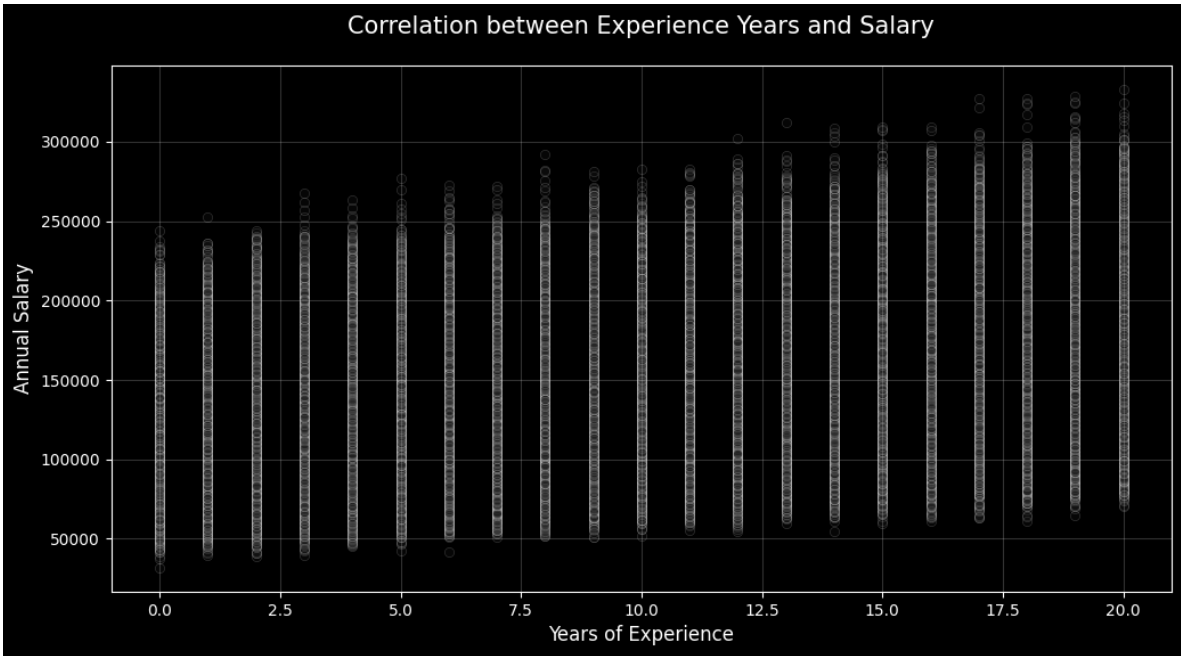
Example: We might have a country with a high average salary where most of the data pertains to high school diplomas, while a country with a low average salary where most of the data pertains to doctorates. This might not apply to these data points, but it's just an example.

II. Exploratory Data Analysis (EDA)

II-1. Correlation Analysis: Years of Experience vs. Annual Salary

I-1-1. Comprehensive Salary Mapping: A 250K-Record Scatter Analysis by years of experience

In [17]:
```
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))

sns.scatterplot(data=df, x='experience_years', y='salary', alpha=0.2, col

plt.title('Correlation between Experience Years and Salary', fontsize=15,
plt.xlabel('Years of Experience', fontsize=12)
plt.ylabel('Annual Salary', fontsize=12)

plt.grid(True, alpha=0.2)
plt.show()
```

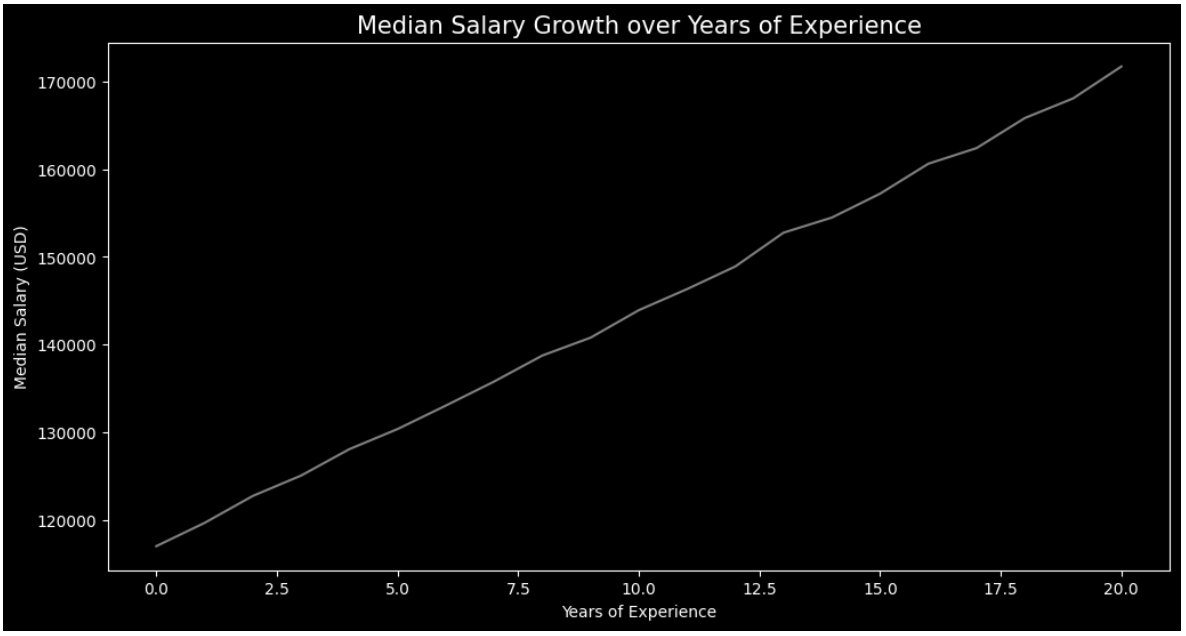


Based on the visualized 250,000 records in Correlation between Experience Years and Salary scatter plot, we can observe the following:

- Upward Trend in Max and Min Salaries: There is a clear positive correlation where an increase in Years of Experience leads to a consistent rise in both the Minimum and Maximum salary levels within the dataset.
- Shift in Density Concentration: While the overall range expands, we observe that the highest density of points (the darkest areas in the plot) gradually shifts upward as experience increases. This indicates that the "market standard" for a typical salary moves to a higher baseline with seniority.
- Salary Overlap: The data shows that certain individuals with fewer years of experience earn higher salaries than others with significantly more seniority, suggesting that other factors (skills,industry or country) play a major role.
- Next Step: To gain a clearer understanding of the typical career progression, we will conduct a comparison of Median Salaries across all years of experience in the data.

II-1-2. Median Salary Growth over Years of Experience

In [18]:
```
plt.style.use('dark_background')
plt.figure(figsize=(12, 6))

sns.lineplot(data=df, x='experience_years', y='salary', estimator=np.medi

plt.title('Median Salary Growth over Years of Experience', fontsize=15)
plt.xlabel('Years of Experience')
plt.ylabel('Median Salary (USD)')

plt.show()
```



Based on the Median Salary Growth over Years of Experience plot, we can observe the following:

```
Consistent Median Increase: There is a clear and steady
upward trend in the median salary as years of experience
increase.
```

II-1-3. Year-over-Year Salary Growth: Absolute and Percentage Analysis

In [19]:
```
median_salaries = df.groupby('experience_years')['salary'].median()

distinct_experience = sorted(df['experience_years'].unique())
print("Distinct Experience Years:", distinct_experience)

print("-" *30)

is_consecutive = (median_salaries.index.to_series().diff().dropna() == 1)
print(f"Are years consecutive? {is_consecutive}")

print("-" * 30)

print(df['experience_years'].value_counts().sort_index())
```

```
Distinct Experience Years: [np.int64(0), np.int64(1), np.int64(2), np.int6
4(3), np.int64(4), np.int64(5), np.int64(6), np.int64(7), np.int64(8), n
p.int64(9), np.int64(10), np.int64(11), np.int64(12), np.int64(13), np.int
64(14), np.int64(15), np.int64(16), np.int64(17), np.int64(18), np.int64(1
9), np.int64(20)]
------------------------------
Are years consecutive? True
------------------------------
experience_years
0     11804
1     12011
2     11950
3     11890
4     12001
5     11862
6     11860
7     11981
8     11822
9     11856
10    11777
11    11747
12    11944
13    11699
14    12166
15    12052
16    11908
17    11735
18    12102
19    11875
20    11958
Name: count, dtype: int64
```

Before proceeding with the Year-over-Year growth analysis, it is crucial to validate the structure and volume of our data to ensure the mathematical results are both accurate and representative.

1. Why check for Consecutive Years?

   Mathematical Accuracy: To calculate the Yearly Increase and Growth Rate (%) using methods like .diff() and .pct_change(), the years of experience must be consecutive (e.g., 0, 1, 2...).

   The Logic: If a year is missing (e.g., jumping from 5 to 7 years), the calculation would incorrectly show a 2-year jump as a 1-year annual increase. By confirming that is_consecutive is True, we ensure our growth metrics reflect actual annual changes.

2. Why check Value Counts per Year?

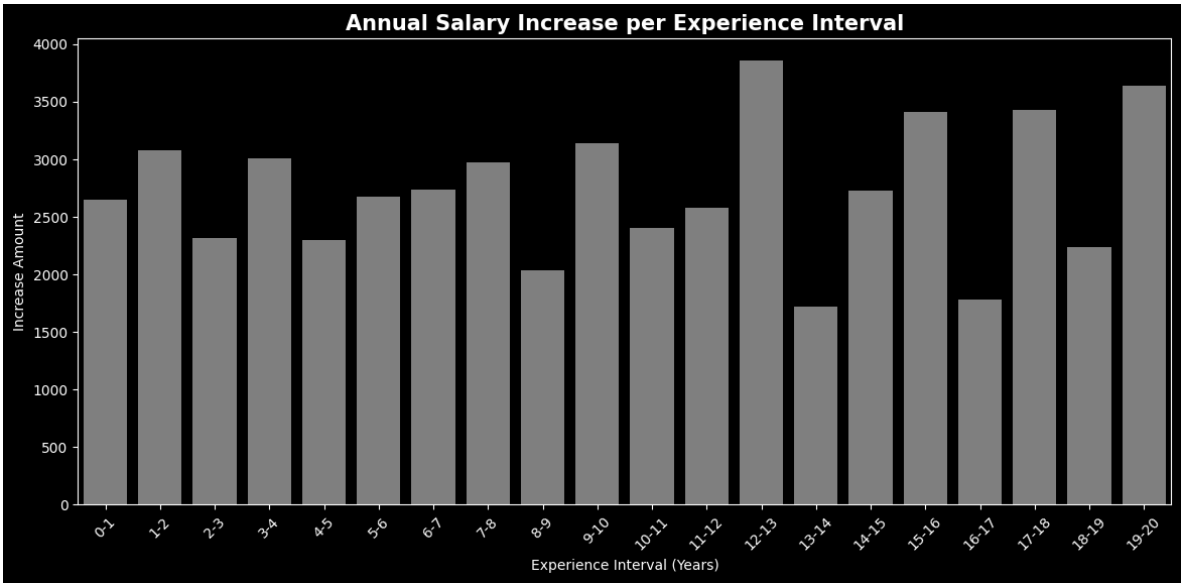
   Statistical Significance: With a massive dataset of 250,000 records, we need to ensure that each "year of experience" category has a sufficient number of observations.

   Reliability of the Median: A median calculated from 10,000 people is much more "stable" and "trustworthy" than a median calculated from only 10 or 20 people. Checking the value_counts() allows us to identify if certain seniority levels have enough data to be considered a reliable market standard.

In [20]:
```
yearly_increase = median_salaries.diff()
percentage_growth = median_salaries.pct_change() * 100
intervals = [f"{int(i-1)}-{int(i)}" for i in median_salaries.index[1:]]

growth_report = pd.DataFrame({
    'Experience Gap': intervals,
    'Annual Increase': yearly_increase.iloc[1:].values,
    'Growth Rate (%)': percentage_growth.iloc[1:].values
})

print(growth_report)
```

```
   Experience Gap  Annual Increase  Growth Rate (%)
0             0-1           2652.5         2.266774
1             1-2           3075.5         2.570006
2             2-3           2319.0         1.889290
3             3-4           3012.5         2.408776
4             4-5           2295.5         1.792295
5             5-6           2676.0         2.052596
6             6-7           2739.5         2.059039
7             7-8           2969.0         2.186513
8             8-9           2038.0         1.468765
9            9-10           3136.0         2.227368
10          10-11           2404.0         1.670256
11          11-12           2581.0         1.763773
12          12-13           3856.0         2.589397
13          13-14           1716.0         1.123250
14          14-15           2727.5         1.765521
15          15-16           3414.5         2.171873
16          16-17           1784.0         1.110634
17          17-18           3426.5         2.109745
18          18-19           2237.5         1.349196
19          19-20           3640.0         2.165674
```

In [21]:
```
plt.figure(figsize=(14, 6))

sns.barplot(data=growth_report, x='Experience Gap', y='Annual Increase',

plt.title('Annual Salary Increase per Experience Interval', fontsize=15,
plt.xlabel('Experience Interval (Years)')
plt.ylabel('Increase Amount')
plt.xticks(rotation=45)
plt.show()
```

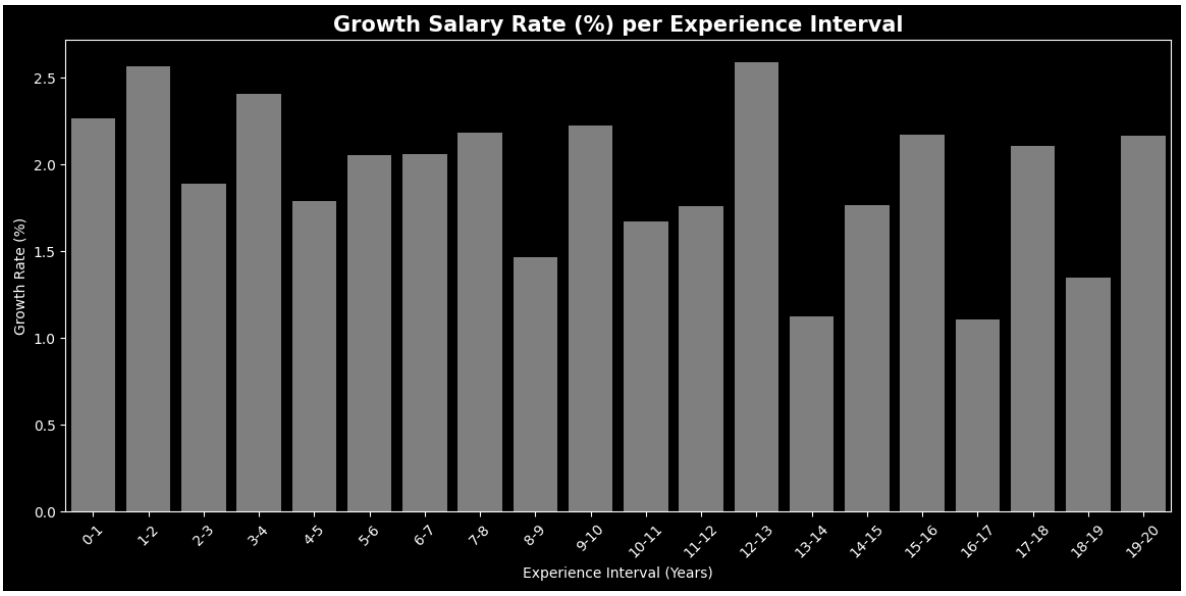


From Annual Salary Increase ($) per Experience Interval bar chart:

Volatility in Growth: The annual salary increase does not follow a predictable linear path, showing significant fluctuation between years.

In [22]:
```
plt.figure(figsize=(14, 6))

sns.barplot(data=growth_report, x='Experience Gap', y='Growth Rate (%)',c
```

```
plt.title('Growth Salary Rate (%) per Experience Interval', fontsize=15,
plt.xlabel('Experience Interval (Years)')
plt.ylabel('Growth Rate (%)')
plt.xticks(rotation=45)
plt.show()
```



Volatility in Growth:

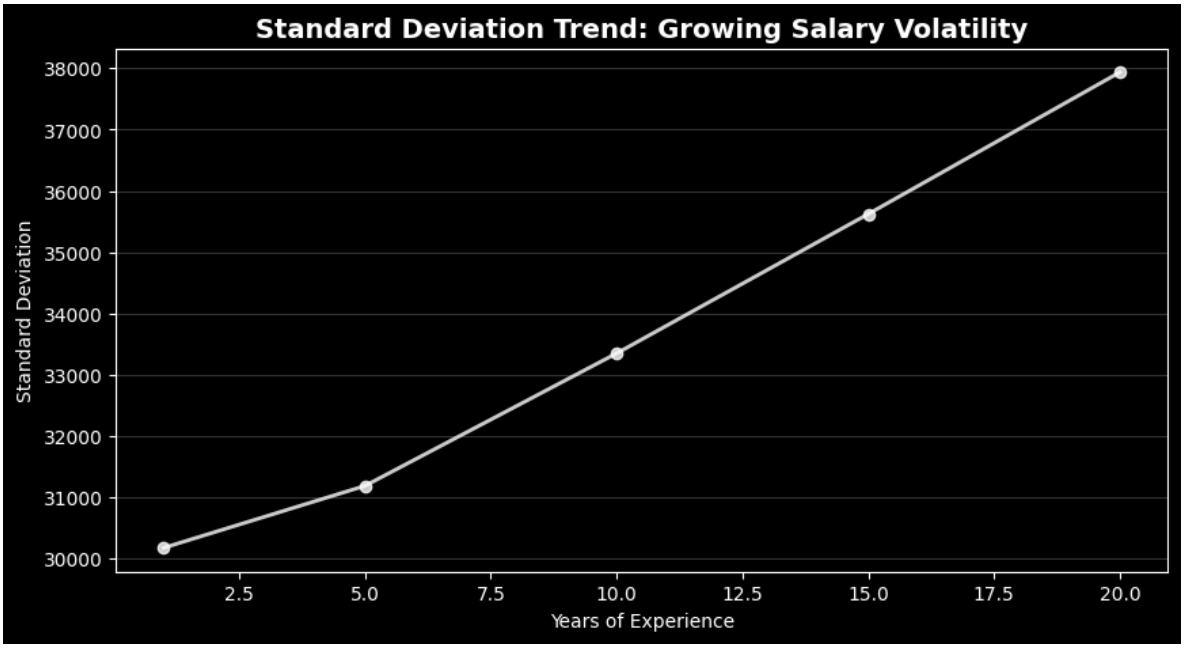
The annual salary growth rate does not follow a predictable linear path, showing significant fluctuations across experience intervals.

II-1-4. Standard Deviation Trend: Growing Salary Volatility

In [23]:
```
std_by_year = df.groupby('experience_years')['salary'].std()

target_years = [1, 5, 10, 15, 20]
target_std = std_by_year.loc[target_years]

std_report = target_std.reset_index()
std_report.columns = ['Years of Experience', 'Standard Deviation']
print(std_report)
```

```
   Years of Experience  Standard Deviation
0                    1        30178.353663
1                    5        31189.351867
2                   10        33349.987461
3                   15        35622.615018
4                   20        37934.873519
```

In [24]:
```
plt.figure(figsize=(10, 5))
plt.plot(std_report['Years of Experience'], std_report['Standard Deviatio
         marker='o', color='#fff', alpha=0.8, linewidth=2)

plt.title('Standard Deviation Trend: Growing Salary Volatility', fontsize
plt.xlabel('Years of Experience')
plt.ylabel('Standard Deviation')
plt.grid(axis='y', alpha=0.2)
plt.show()
```



Technical Analysis: Standard Deviation as an Indicator of Market Volatility

Linear Development of Risk: Standard deviation follows a consistently upward linear path. This indicates a structural shift in the market as professionals gain more experience.

Prediction vs. Variance: In the early stages of a career (first year), a low standard deviation (around 30,000) suggests a more uniform salary scale compared to those with more years of experience. However, by year 20, the variance widens to approximately $38,000, demonstrating that more years of experience lead to greater salary variance.

Results Interpretation: From Standardization to Divergence

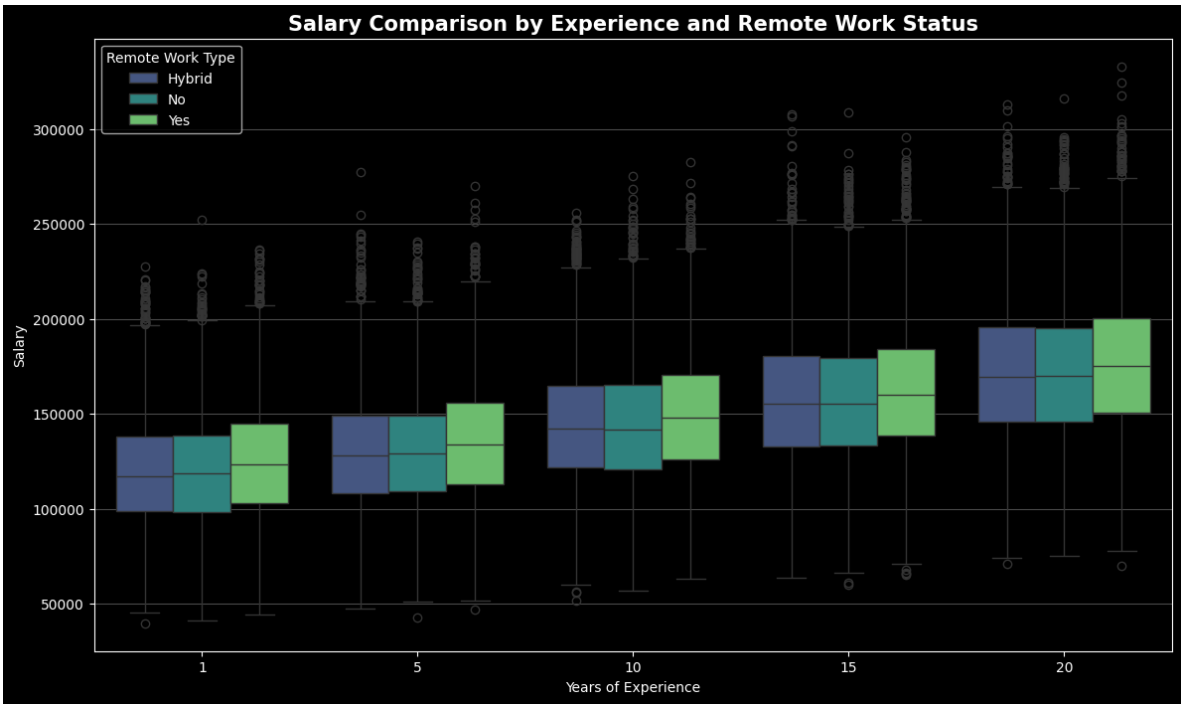
The linear growth of Standard Deviation alongside increasing Salary Variance leads to several key insights:

- As careers mature, Seniority does not guarantee a high salary; rather, it expands the potential range, meaning the gap between top performers and the median wider significantly over time.
- Market Maturity: The rising variance proves that the industry rewards long-term specialization disproportionately. Tenure acts as a "multiplier" for those who successfully position themselves in high-demand sub-sectors, creating the "Outliers" seen in the 300k+ range.

II-1-5. Salary Comparison by Experience and Remote Work Status

In [25]:
```
years_to_show = [1, 5, 10, 15, 20]
df_filtered = df[df['experience_years'].isin(years_to_show)]

plt.figure(figsize=(14, 8))
sns.boxplot(data=df_filtered, x='experience_years', y='salary', hue='remo
            palette='viridis', showfliers=True)
```

```
plt.title('Salary Comparison by Experience and Remote Work Status', fonts
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend(title='Remote Work Type')
plt.grid(axis='y', alpha=0.3)
plt.show()
```



Analysis: The Impact of Remote Work on Salary Trajectory

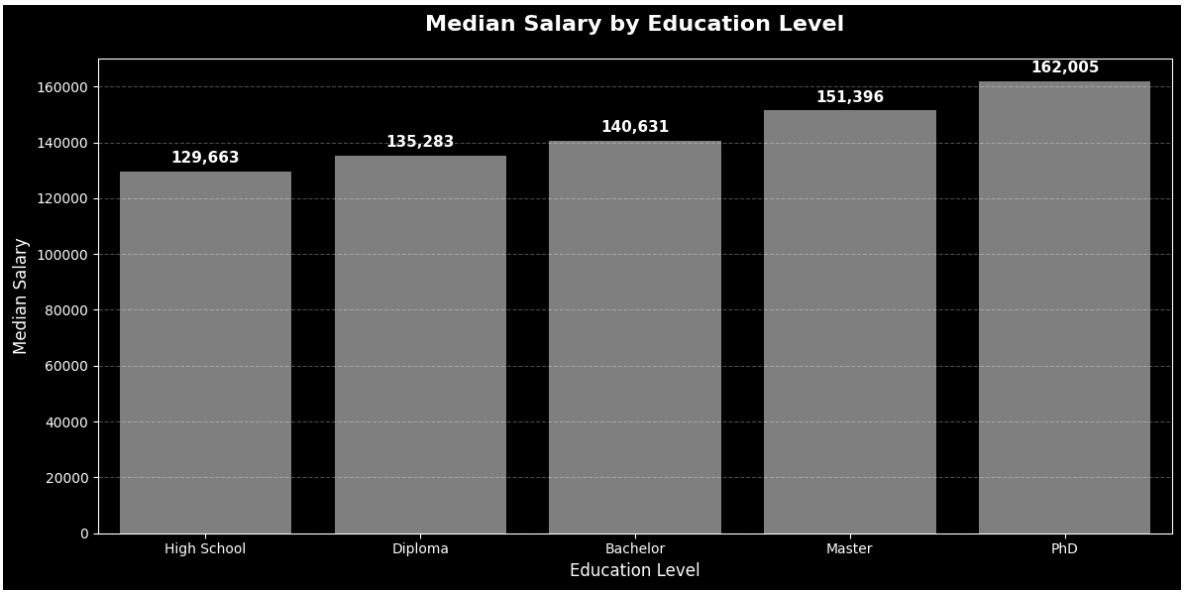
```
Remote Work Advantage: Across all experience levels, fully
remote roles  consistently outperform Hybrid and In-person
roles in terms of median salary.


Market Standard: Hybrid and Non-remote roles show similar
salary distributions, suggesting that the primary market
distinction is between "Fully Remote" vs. "Office-
Affiliated" roles.
```

II-2. Analyzing the Correlation of Education Level with Salary

II-2-1. Median Salary by Education Level

In [26]:
```
education_order = ['High School', 'Diploma', 'Bachelor', 'Master', 'PhD']

plt.figure(figsize=(12, 6))

sns.barplot(
    data=df,
    x='education_level',
    y='salary',
    order=education_order,
    estimator='median',
    errorbar=None,
    color="#fff",
```

```
    alpha=0.5
)

plt.title('Median Salary by Education Level', fontsize=16, fontweight='bo
plt.xlabel('Education Level', fontsize=12)
plt.ylabel('Median Salary', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.3)

medians = df.groupby('education_level')['salary'].median().loc[education_
for i, value in enumerate(medians):
    plt.text(i, value + 2000, f'{int(value):,}', ha='center', va='bottom'


plt.tight_layout()
plt.show()
```



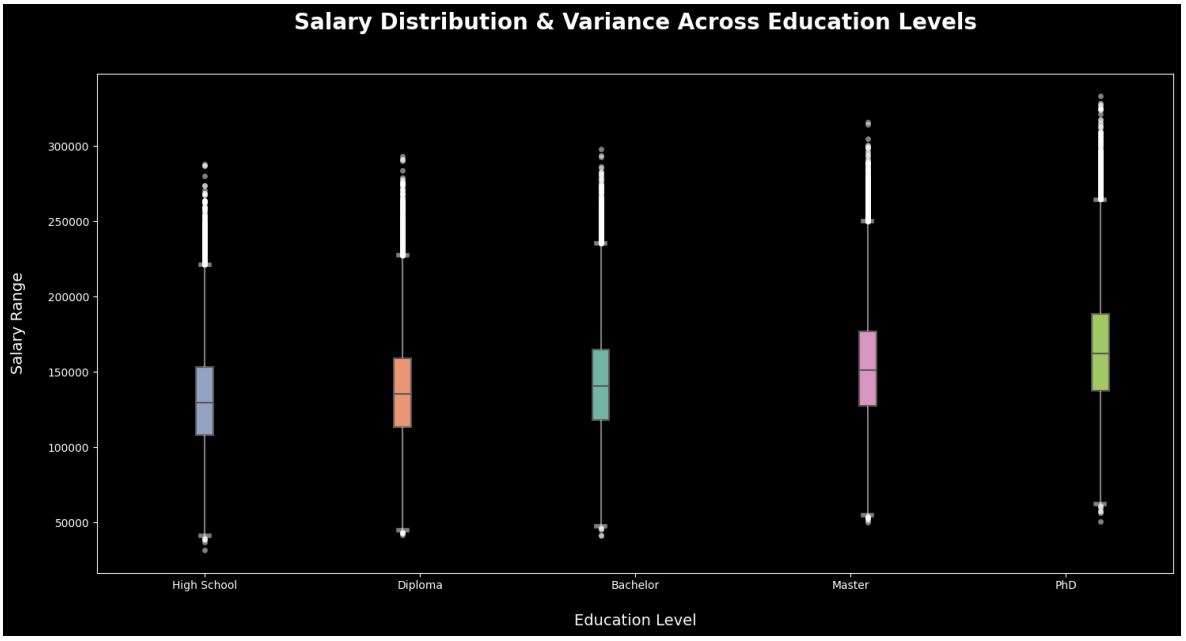
Consistent Growth:

- There is a clear linear progression where higher education levels directly correlate with increased median salaries.
- PhD Dominance: PhD holders command the highest median salary (162,005)
- Educational Floor: The relatively high baseline for High School graduates indicates a lucrative industry

II-2-2. Salary Distribution & Variance Across Education Levels

In [27]:
```
plt.style.use('dark_background')

fig, ax = plt.subplots(figsize=(16, 9))

education_order = ['High School', 'Diploma', 'Bachelor', 'Master', 'PhD']

sns.boxplot(
    data=df,
    x='education_level',
    y='salary',
    order=education_order,
    hue='education_level',
    palette='Set2',
    legend=False,
```

```
    linewidth=1.5,
    width=0.4,
    ax=ax,
    flierprops={
        'marker': 'o',
        'markersize': 5,
        'markerfacecolor': 'auto',
        'markeredgecolor': 'none',
        'alpha': 0.5
    },
    whiskerprops={'color': '#ffffff', 'linewidth': 1.5, 'alpha': 0.5},
    capprops={'color': '#ffffff', 'linewidth': 4, 'alpha': 0.5}
)

ax.set_title('Salary Distribution & Variance Across Education Levels',
             fontsize=20, fontweight='bold', pad=40)

ax.set_xlabel('Education Level', fontsize=14, labelpad=20)
ax.set_ylabel('Salary Range', fontsize=14, labelpad=20)


plt.subplots_adjust(top=0.85, bottom=0.15, left=0.1, right=0.95)

plt.show()
```



This box chart provides a comprehensive statistical overview of how wages relate to academic qualifications. It reveals the full picture beyond simple averages, focusing on volatility, outliers, and the minimum income threshold.

1. **The “Minimum Income” Effect**

There is a clear direct correlation between educational level and minimum wage.

The bottom lines of the rectangle (representing the lowest 25% of earners) rise steadily from high school to doctorate.

2. **Significant Overlap Between Quartiles**

Despite the high median, there is significant overlap between the boxes (the middle half

of the data).

A high school graduate can earn more than someone with a doctorate for several reasons.

Academic qualifications are not the sole determinant of salary. Factors such as specialized skills, years of experience, country, and others also influence pay.

II-2-3. The Interplay between Education & Experience on Compensation

In [28]:
```
pivot_df = df.pivot_table(
    index='education_level',
    columns='experience_years',
    values='salary',
    aggfunc='median'
).reindex(['High School', 'Diploma', 'Bachelor', 'Master', 'PhD'])
```

In [29]:
```
pivot_df
```

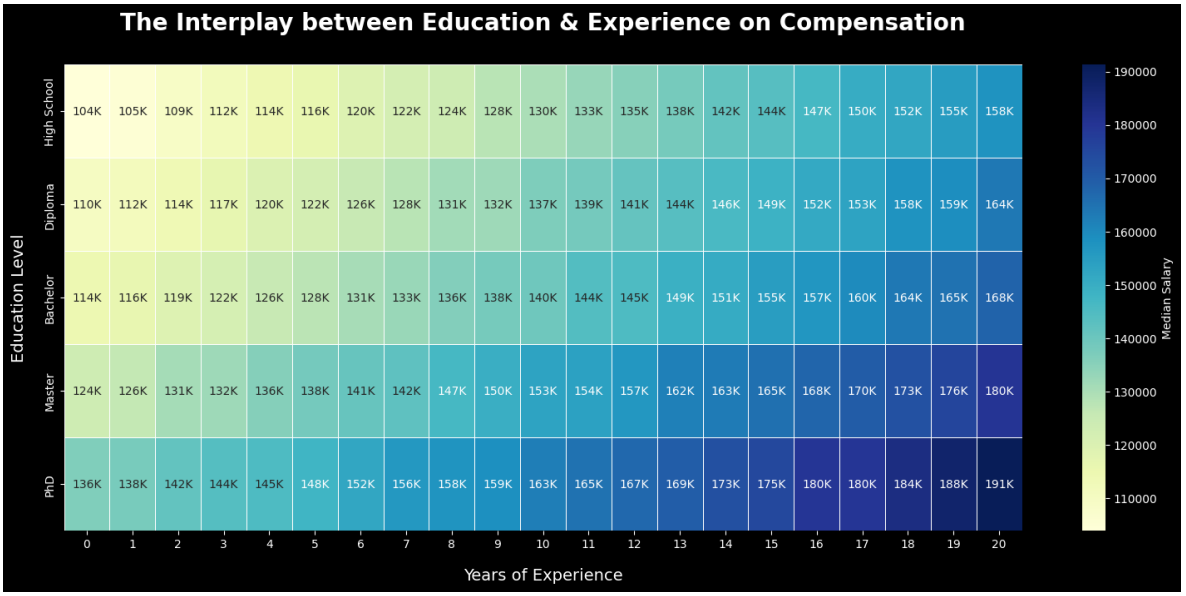
Out[29]:

experience_years	0	1	2	3	4	5	
education_level							
High School	103883.0	105300.5	109081.0	111767.0	114254.0	115870.5	119501.
Diploma	110234.0	111650.0	114102.5	116739.5	119975.0	122045.5	125699.
Bachelor	114488.0	116437.0	119379.0	121898.5	125598.0	127560.0	130586.
Master	123970.0	126304.0	131129.5	131908.0	135940.0	138488.0	140984.
PhD	136343.5	138009.0	141528.0	143979.0	145207.0	147817.0	152277.

5 rows × 21 columns

In [30]:
```
plt.style.use('dark_background')

annot_text = pivot_df.map(lambda x: f'{x/1000:.0f}K' if pd.notnull(x) els

fig, ax = plt.subplots(figsize=(18, 8))

sns.heatmap(
    pivot_df,
    annot=annot_text,
    fmt="",
    cmap="YlGnBu",
    linewidths=.5,
    cbar_kws={'label': 'Median Salary'},
    ax=ax
)

ax.set_title('The Interplay between Education & Experience on Compensatio
             fontsize=20, fontweight='bold', pad=30)

ax.set_xlabel('Years of Experience', fontsize=14, labelpad=15)
ax.set_ylabel('Education Level', fontsize=14, labelpad=15)

plt.subplots_adjust(top=0.85, bottom=0.15, left=0.15, right=0.95)
```

```
plt.show()
```



Experience as a Key Factor

The horizontal color gradient confirms that years of experience consistently contribute to salary growth at all educational levels. Even without a college degree, 20 years of experience can increase a high school graduate's salary by approximately 52% (from $104,000 to $158,000).

The Trade-off Between Degree and Experience

The heat map demonstrates that professional experience effectively bridges the salary gap between academic qualifications and academic qualifications. For example, a professional with a bachelor's degree and 10 years of experience earns $140,000 more than someone with a recent master's degree ($124,000–$126,000). This shows that practical experience is highly valued and can compensate for lower academic qualifications over time.

The “Master’s and PhD Advantage”

Advanced degrees offer a “better start” and faster career progression. While it takes a bachelor's degree holder twenty years to earn a salary of $168,000, a PhD holder reaches the same level within 12 years of experience.

II-3. Country-wise Comparison

II-3.1. Statistical Summary: Median Salary by Country

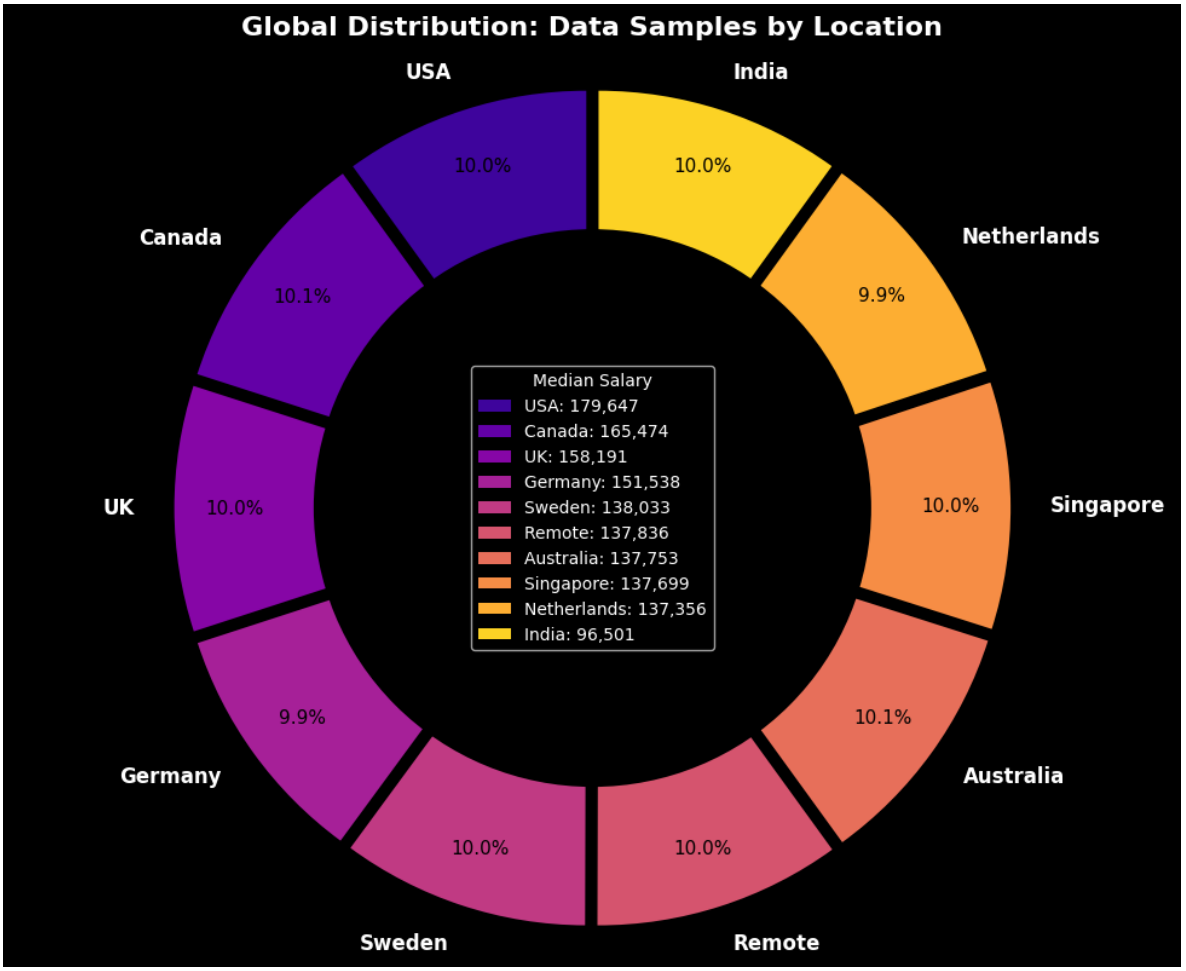
In [31]:
```
country_analysis = df.groupby('location')['salary'].agg(['median', 'count
country_analysis
```

Out[31]:

location	median	count
USA	179647.0	24931
Canada	165474.0	25165
UK	158191.0	24927
Germany	151538.0	24763
Sweden	138033.0	25100
Remote	137836.0	25065
Australia	137753.5	25258
Singapore	137699.0	25035
Netherlands	137356.0	24861
India	96501.0	24895

In [32]:

```
plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 8))

wedges, texts, autotexts = ax.pie(country_analysis['count'],
                                  labels=country_analysis.index,
                                  autopct='%1.1f%%',
                                  startangle=90,
                                  colors=sns.color_palette("plasma", len(
                                  pctdistance=0.85,
                                  explode=[0.05] * len(country_analysis))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold", color="white")
plt.setp(autotexts, size=11, color="black")

ax.axis('equal')
plt.title('Global Distribution: Data Samples by Location', fontsize=16, f

salary_info = [f"{loc}: {int(med):,}" for loc, med in zip(country_analysi
ax.legend(wedges, salary_info, title="Median Salary", loc="center", fonts

plt.tight_layout()
plt.show()
```

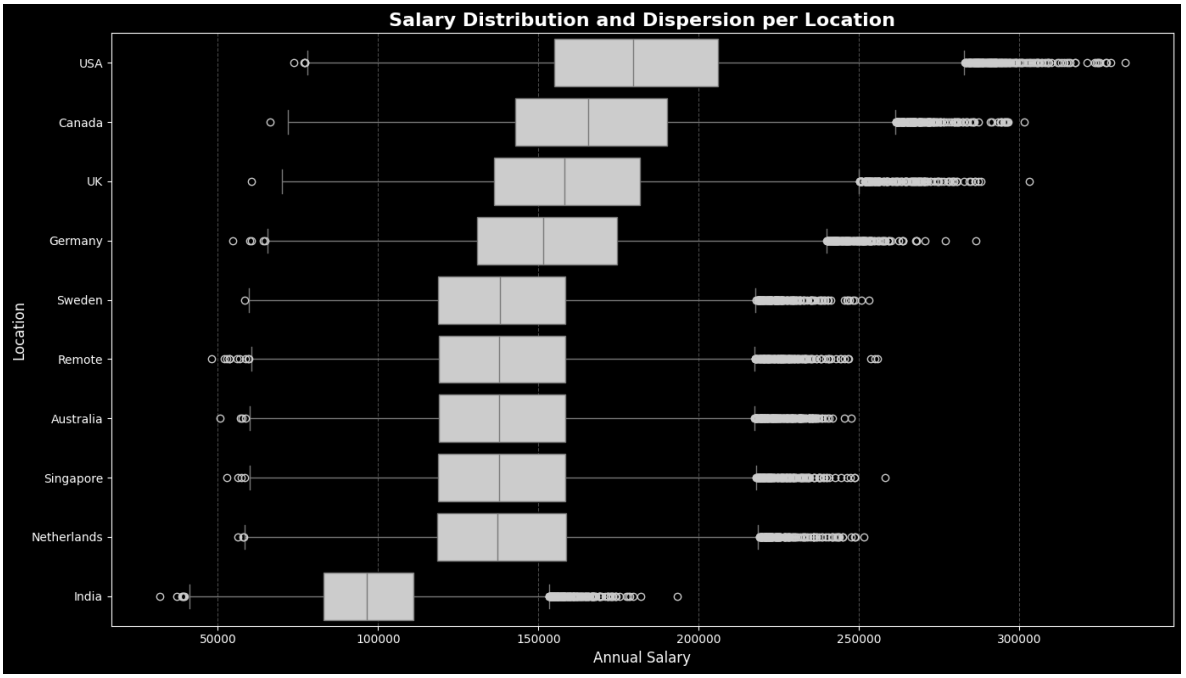


A homogeneous distribution: The graph shows a nearly uniform distribution of data samples across all locations (approximately 10% per location).

Salary disparities: Despite the equal sample sizes, average salaries vary significantly, with the USA topping the list at 179,647 and India at the bottom at 96,501.

II-3-2. Salary Dispersion: Identifying Income Gaps by Location

In [33]:
```
plt.figure(figsize=(14, 8))
plt.style.use('dark_background')

sns.boxplot(x='salary', y='location', data=df,
            order=country_analysis.index,
            flierprops={"marker": "o", "markeredgecolor": "#ccc"},
           color="#ccc"            )

plt.title('Salary Distribution and Dispersion per Location', fontsize=16,
plt.xlabel('Annual Salary', fontsize=12)
plt.ylabel('Location', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)

plt.tight_layout()
plt.show()
```



In [34]:
```
stats_table = df.groupby('location')['salary'].agg([
    ('Median', 'median'),
    ('25%', lambda x: x.quantile(0.25)),
    ('75%', lambda x: x.quantile(0.75))
])

stats_table['IQR (Difference)'] = stats_table['75%'] - stats_table['25%']

stats_table = stats_table.sort_values(by='Median', ascending=False)

styled_stats = stats_table.style.format("{:,.0f}")
styled_stats
```

Out[34]:

location	Median	25%	75%	IQR (Difference)
USA	179,647	154,842	206,070	51,228
Canada	165,474	142,659	190,110	47,451
UK	158,191	136,178	181,744	45,567
Germany	151,538	130,754	174,424	43,670
Sweden	138,033	118,730	158,346	39,616
Remote	137,836	118,931	158,410	39,479
Australia	137,754	118,978	158,353	39,375
Singapore	137,699	118,626	158,334	39,708
Netherlands	137,356	118,479	158,621	40,142
India	96,501	83,018	111,158	28,140

Detailed Notes:

Mechanical Correlation Between Ranges: There is an unusual correlation between the minimum, maximum, and median values. In real-world datasets, a country with a low

median might have very high wages (e.g., few high-income earners), but here, no region with a low median exceeds the maximum wage in countries with the highest median. This suggests that the data may be artificially generated and not based on a naturally occurring random sample.

Mirror Effect (Artificial Symmetry): There is a near-perfect and suspiciously similar match between the samples from Sweden, Australia, Singapore, the Outback, and the Netherlands. These four categories are almost identical across five different statistical indicators:

- Minimum and maximum values.
- Meaning.
- 25th and 75th percentiles (quartiles).

This level of mathematical symmetry across diverse geographical locations strongly supports the hypothesis that this dataset may be artificial.

Interquartile Range (IQR) Analysis: * USA: Shows the largest difference between the 75th and 25th percentiles (interquartile range), indicating the most flexible salary distribution in this dataset.

India: Shows the smallest difference (interquartile range), indicating a very narrow salary range that lacks the diversity typical of a real labor market.

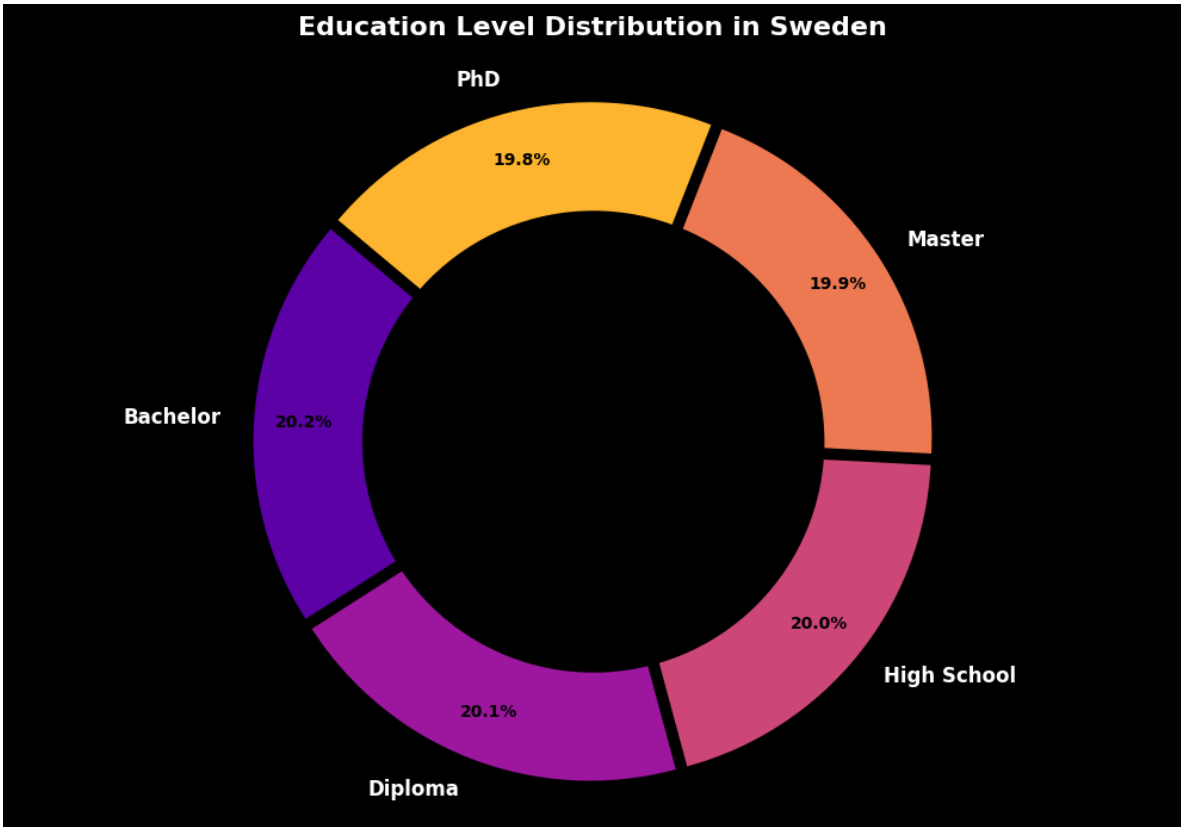
Media-Dispersion Relationship: We observe a direct, positive relationship between the median and the interquartile range. As the median increases, the gap between the 75th and 25th percentiles increases proportionally.

II-3-3. Data Samples by Education Level

In [35]:
```
edu_loc_table = pd.crosstab(df['location'], df['education_level'])
display(edu_loc_table)

sweden_data = edu_loc_table.loc['Sweden']

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 7))

wedges, texts, autotexts = ax.pie(sweden_data,
                                  labels=sweden_data.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("plasma", len(
                                  pctdistance=0.85,
                                  explode=[0.03] * len(sweden_data))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold")
plt.setp(autotexts, size=10, color="black", weight="bold")
```

```
plt.title('Education Level Distribution in Sweden', fontsize=16, fontweig
ax.axis('equal')
plt.tight_layout()
plt.show()
```

education_level location	Bachelor	Diploma	High School	Master	PhD
Australia	4983	4997	5169	5102	5007
Canada	5025	4997	4952	5160	5031
Germany	4868	4927	5053	4958	4957
India	5109	4922	4932	4956	4976
Netherlands	4912	4922	5041	5009	4977
Remote	4990	4973	4993	5095	5014
Singapore	5069	5053	4935	4996	4982
Sweden	5065	5052	5013	4995	4975
UK	4997	5010	4990	4974	4956
USA	4932	4923	4987	5107	4982



Observation and Analysis:

The "Constant Number" Phenomenon: When examining the distribution of educational levels in different geographical locations (such as Sweden, the USA, and India), we observe a remarkable similarity in the number of samples for each category.

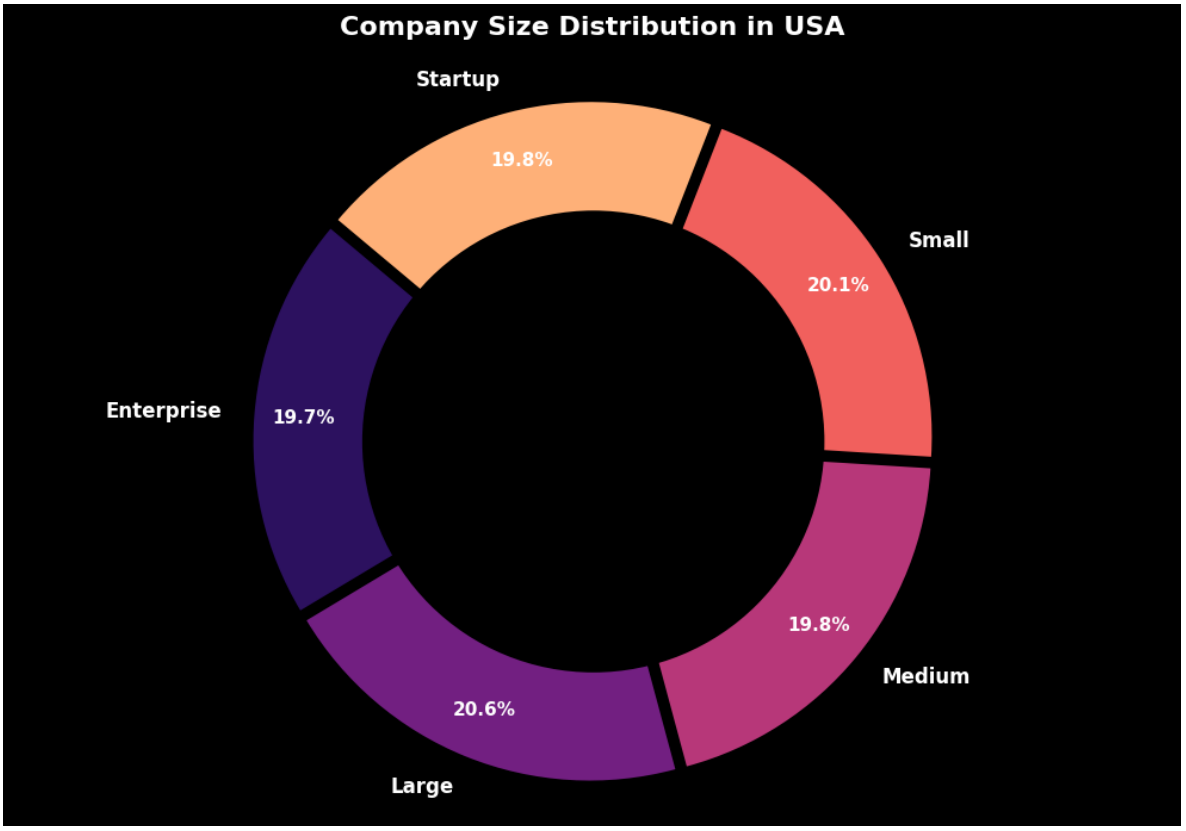
The 20% Uniformity Rule: In each country, each educational level (high school, bachelor's degree, etc.) represents approximately 20% of the total sample.

This near-perfect similarity strongly supports the hypothesis of artificial data.

II-3-4. Corporate Landscape: Company Size Distribution by Location

In [36]:
```
company_loc_table = pd.crosstab(df['location'], df['company_size'])
display(company_loc_table)

usa_company_data = company_loc_table.loc['USA']

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 7))

wedges, texts, autotexts = ax.pie(usa_company_data,
                                  labels=usa_company_data.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("magma", len(u
                                  pctdistance=0.85,
                                  explode=[0.03] * len(usa_company_data))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold")
plt.setp(autotexts, size=11, color="white", weight="bold")

plt.title('Company Size Distribution in USA', fontsize=16, fontweight='bo
ax.axis('equal')
plt.tight_layout()
plt.show()
```

company_size location	Enterprise	Large	Medium	Small	Startup
Australia	4988	5050	5108	5135	4977
Canada	5016	4913	5058	5055	5123
Germany	4850	5092	4943	4993	4885
India	5002	4975	4985	4934	4999
Netherlands	5027	4967	5071	4920	4876
Remote	5059	5044	4832	5084	5046
Singapore	4966	5024	5007	5101	4937
Sweden	5080	5085	5037	4948	4950
UK	4973	4973	5043	5061	4877
USA	4914	5131	4943	5004	4939



Observation and Analysis:

The **"Constant Number"** Phenomenon: When examining the distribution of company sizes in different geographical locations (such as Sweden, the USA, and India), we observe a remarkable similarity in the number of samples for each category.

The 20% Uniformity Rule: In each country, each company size represents approximately 20% of the total sample.

Conclusion: This near-perfect similarity strongly supports the hypothesis of artificial data.

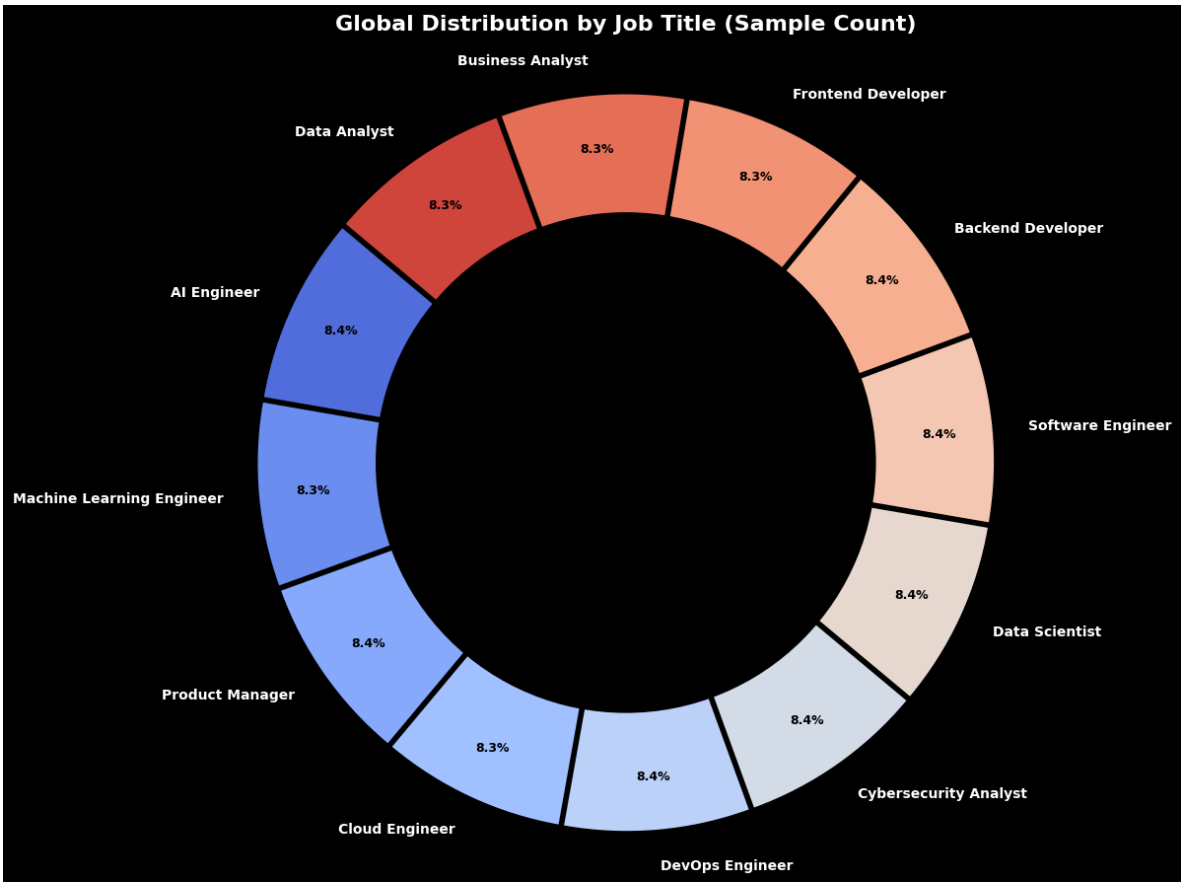
II-4. Job title comparison

II-4-1. Statistical Summary: Median Salary by Job Title

In [37]:
```
job_stats = df.groupby('job_title')['salary'].agg(['median', 'count']).so
display(job_stats)

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(12, 9))

wedges, texts, autotexts = ax.pie(job_stats['count'],
                                  labels=job_stats.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("coolwarm", le
                                  pctdistance=0.85,
                                  explode=[0.03] * len(job_stats))

centre_circle = plt.Circle((0,0), 0.70, fc='black')
```

```
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=10, weight="bold")
plt.setp(autotexts, size=9, color="black", weight="bold")

plt.title('Global Distribution by Job Title (Sample Count)', fontsize=16,
ax.axis('equal')
plt.tight_layout()
plt.show()
```

job_title	median	count
AI Engineer	171674.0	20945
Machine Learning Engineer	160933.0	20677
Product Manager	155655.5	20950
Cloud Engineer	150005.0	20686
DevOps Engineer	148021.0	20889
Cybersecurity Analyst	146737.0	20959
Data Scientist	145231.5	20890
Software Engineer	139432.5	20876
Backend Developer	136678.0	21125
Frontend Developer	130619.0	20633
Business Analyst	120562.0	20648
Data Analyst	118079.5	20722



The "Constant Number" Phenomenon: When examining the distribution of the 12 main job titles (such as Data Scientist, ML Engineer, Data Analyst, etc.), we observe a remarkable similarity in the number of samples for each category.

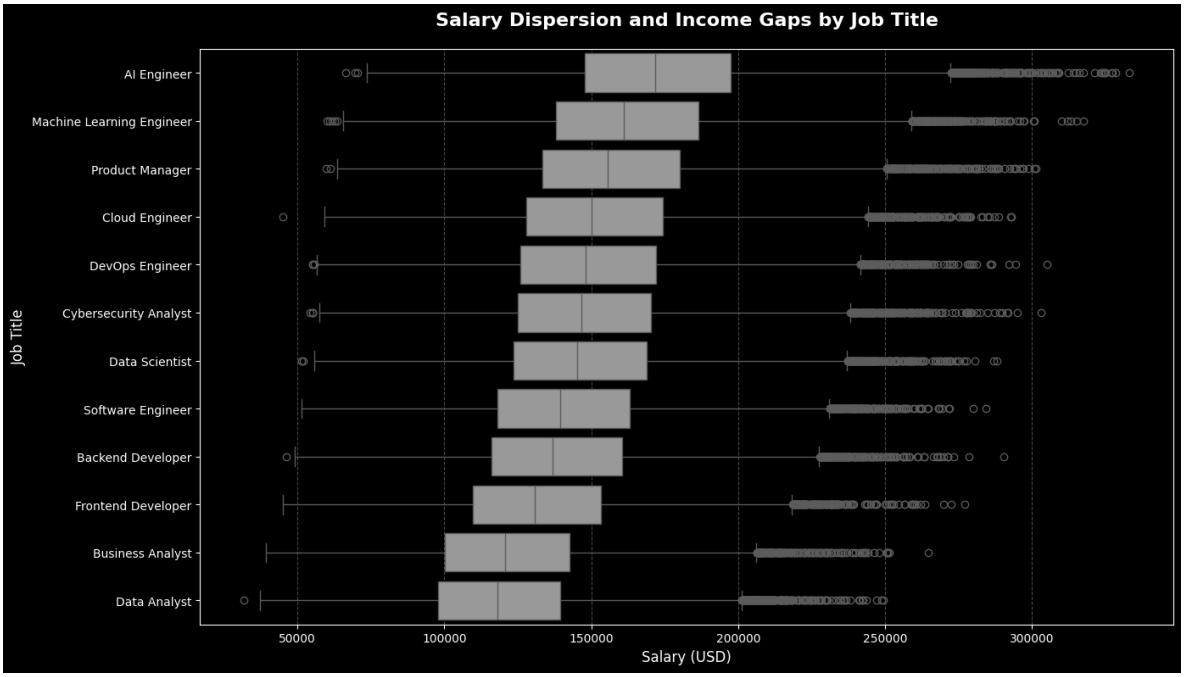
The 8.3% Uniformity Rule: Within the dataset, each of the 12 job titles represents approximately 8.3% of the total sample (100/12≈8.3%). In a real-world labor market.

Conclusion: This near-perfect similarity across 12 distinct categories strongly supports the hypothesis of artificial data.

II-4-2. Salary Dispersion: Identifying Income Gaps by Job

In [38]:
```
job_dispersion = df.groupby('job_title')['salary'].agg(['min', 'max', 'me
display(job_dispersion)

plt.style.use('dark_background')
plt.figure(figsize=(14, 8))

sns.boxplot(x='salary', y='job_title', data=df, order=job_dispersion.inde

plt.title('Salary Dispersion and Income Gaps by Job Title', fontsize=16,
plt.xlabel('Salary (USD)', fontsize=12)
plt.ylabel('Job Title', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```

job_title	min	max	median	IQR
AI Engineer	66371	333046	171674.0	49835.00
Machine Learning Engineer	59925	317377	160933.0	48416.00
Product Manager	59796	301294	155655.5	46941.00
Cloud Engineer	45030	292855	150005.0	46491.00
DevOps Engineer	54985	304968	148021.0	46239.00
Cybersecurity Analyst	54224	303178	146737.0	45272.50
Data Scientist	51349	288049	145231.5	45421.50
Software Engineer	51313	284442	139432.5	45240.25
Backend Developer	46304	290512	136678.0	44613.00
Frontend Developer	44964	277053	130619.0	43506.00
Business Analyst	39285	264567	120562.0	42368.00
Data Analyst	31867	249354	118079.5	41317.50



In [39]:
```
job_salary_stats = df.groupby('job_title')['salary'].agg([
    ('Median', 'median'),
    ('25%', lambda x: x.quantile(0.25)),
    ('75%', lambda x: x.quantile(0.75)),
    ('IQR (75%-25%)', lambda x: x.quantile(0.75) - x.quantile(0.25))
]).sort_values(by='Median', ascending=False)

display(job_salary_stats)
```

job_title	Median	25%	75%	IQR (75%-25%)
AI Engineer	171674.0	147656.00	197491.00	49835.00
Machine Learning Engineer	160933.0	138044.00	186460.00	48416.00
Product Manager	155655.5	133200.75	180141.75	46941.00
Cloud Engineer	150005.0	127965.00	174456.00	46491.00
DevOps Engineer	148021.0	125860.00	172099.00	46239.00
Cybersecurity Analyst	146737.0	124981.00	170253.50	45272.50
Data Scientist	145231.5	123540.75	168962.25	45421.50
Software Engineer	139432.5	117949.75	163190.00	45240.25
Backend Developer	136678.0	115936.00	160549.00	44613.00
Frontend Developer	130619.0	109773.00	153279.00	43506.00
Business Analyst	120562.0	100276.50	142644.50	42368.00
Data Analyst	118079.5	97952.50	139270.00	41317.50

Observation and Analysis:

Linear Gradient: We observe a consistent and clear correlation, where the minimum and maximum salaries increase proportionally with the median. In a typical job market, some jobs (such as data analyst) may have a high maximum salary due to the presence of

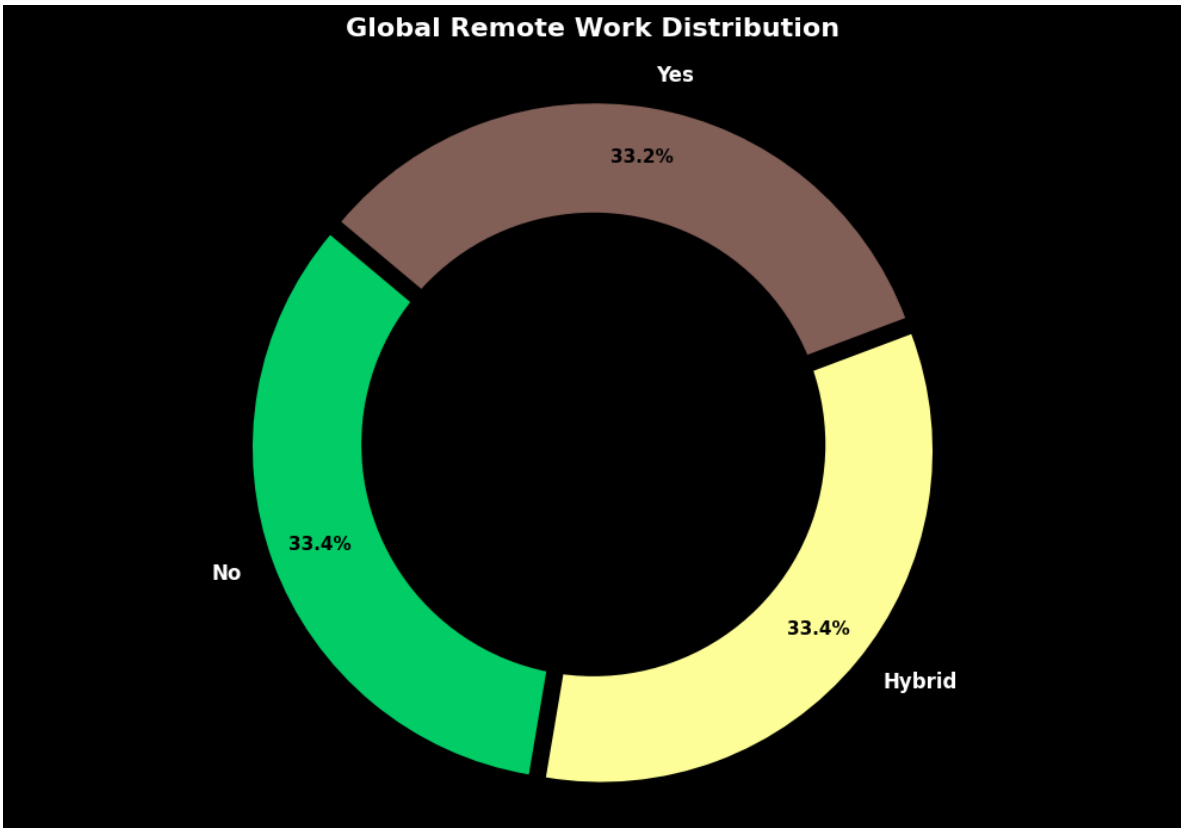
experienced candidates, even if the median is low. However, in this case, the entire range moves as a single, fixed unit.

Relative Dispersion (Interquartile Range): There is a clear and consistent relationship between the median and the interquartile range (75% - 25%). As the average salary for a given job increases, the gap between the 25th and 75th percentiles widens.

II-4-3. Remote Work: Distribution Across Job Titles

In [40]:
```
remote_job_table = pd.crosstab(df['job_title'], df['remote_work'])
display(remote_job_table)

remote_global_dist = df['remote_work'].value_counts()

plt.style.use('dark_background')
fig, ax = plt.subplots(figsize=(10, 7))

wedges, texts, autotexts = ax.pie(remote_global_dist,
                                  labels=remote_global_dist.index,
                                  autopct='%1.1f%%',
                                  startangle=140,
                                  colors=sns.color_palette("terrain", len
                                  pctdistance=0.85,
                                  explode=[0.03] * len(remote_global_dist

centre_circle = plt.Circle((0,0), 0.70, fc='black')
fig.gca().add_artist(centre_circle)

plt.setp(texts, size=12, weight="bold")
plt.setp(autotexts, size=11, color="black", weight="bold")

plt.title('Global Remote Work Distribution', fontsize=16, fontweight='bol
ax.axis('equal')
plt.tight_layout()
plt.show()
```

remote_work job_title	Hybrid	No	Yes
AI Engineer	7049	6902	6994
Backend Developer	7139	6965	7021
Business Analyst	6916	6883	6849
Cloud Engineer	6948	6896	6842
Cybersecurity Analyst	6936	7044	6979
Data Analyst	6829	6987	6906
Data Scientist	7074	6906	6910
DevOps Engineer	6919	7049	6921
Frontend Developer	6880	6922	6831
Machine Learning Engineer	6906	7005	6766
Product Manager	6883	7156	6911
Software Engineer	6997	6906	6973



Observation & Analysis:

The "Constant Number" Phenomenon: When examining the distribution of remote work scenarios (e.g., remote, on-site, hybrid) across the twelve job titles, we observe a remarkable similarity in the number of samples across the entire dataset and also within each job title.

The 33% Rule: Within each job title, each work style represents approximately 1/3 of the samples.

Conclusion: This near-perfect 1/3 distribution across all twelve job titles strongly supports the hypothesis of artificial data.

III. 🤔Is the data synthetic?

In [41]:
```
%%html
<section id="synthetic-data-analysis" style="font-family: 'Segoe UI', Tah

    <h2 style="color: #4db8ff; border-bottom: 2px solid #4db8ff; padding-
        <span style="margin-right: 10px;">🤔</span> Itroduction
    </h2>

    <div class="english-text" style="margin-top: 20px; background-color:
        <p style="font-size: 1.1em; margin: 0;">
            Previously, we presented several pieces of evidence suggestin
        </p>
    </div>



    </div>

</section>
```

🤔 Itroduction

> Previously, we presented several pieces of evidence suggesting that the data might be artificial. However, we will now dedicate a specific section to discuss this topic in detail, using **Job Titles** as a case study to our analysis.

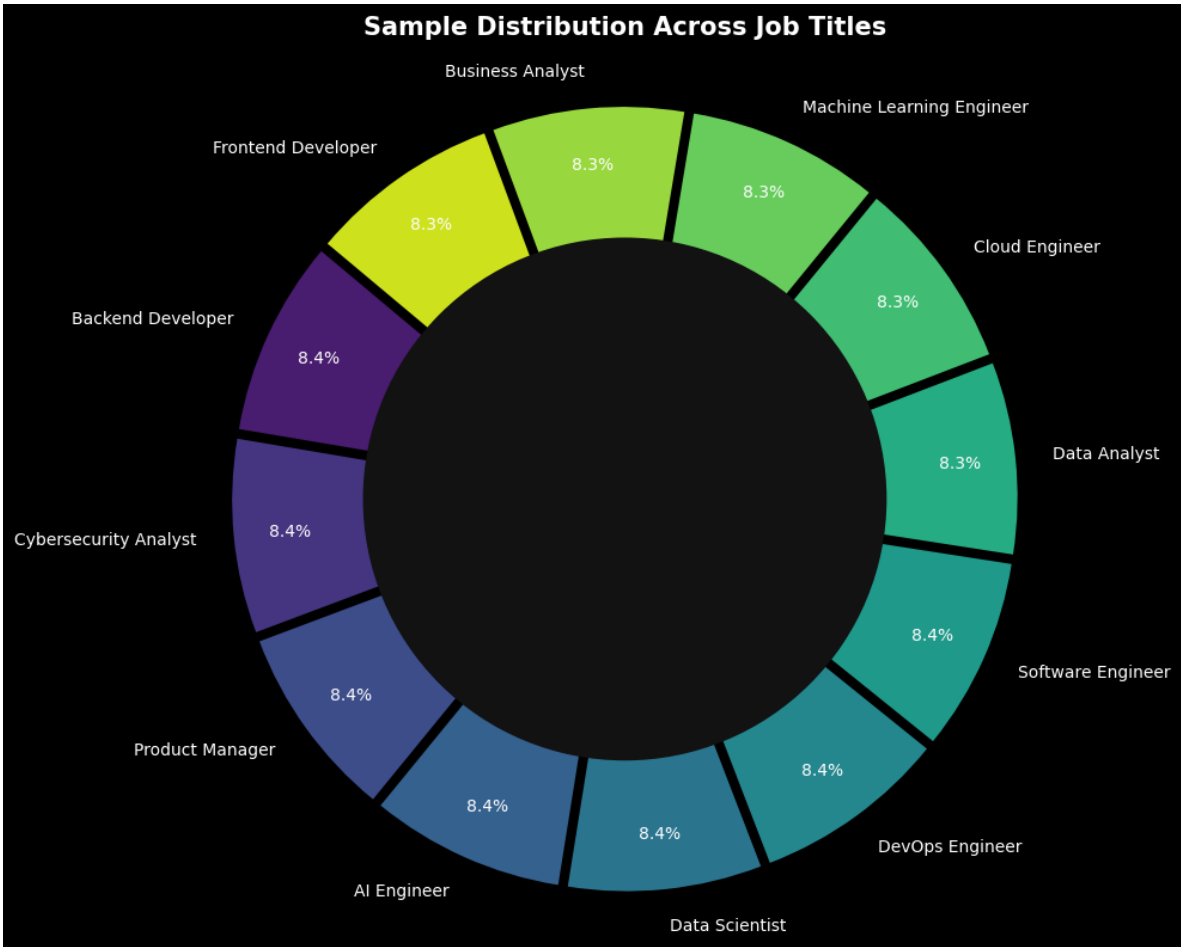
III-1. Statistical Distribution of Sample Sizes Across Job Titles

In [42]:
```
job_counts = df['job_title'].value_counts().reset_index()
job_counts.columns = ['Job Title', 'Sample Count']
job_counts['Percentage (%)'] = (job_counts['Sample Count'] / len(df) * 10

print("Job Title Sample Distribution Table:")
display(job_counts)

plt.style.use('dark_background')
plt.figure(figsize=(10, 8))

colors = sns.color_palette("viridis", len(job_counts))

plt.pie(job_counts['Sample Count'],
        labels=job_counts['Job Title'],
```

```
        autopct='%1.1f%%',
        startangle=140,
        colors=colors,
        pctdistance=0.85,
        explode=[0.05]*len(job_counts))

centre_circle = plt.Circle((0,0), 0.70, fc='#121212')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.title('Sample Distribution Across Job Titles', fontsize=15, fontweigh
plt.axis('equal')
plt.tight_layout()
plt.show()
```

```
Job Title Sample Distribution Table:
```

	Job Title	Sample Count	Percentage (%)
0	Backend Developer	21125	8.45
1	Cybersecurity Analyst	20959	8.38
2	Product Manager	20950	8.38
3	AI Engineer	20945	8.38
4	Data Scientist	20890	8.36
5	DevOps Engineer	20889	8.36
6	Software Engineer	20876	8.35
7	Data Analyst	20722	8.29
8	Cloud Engineer	20686	8.27
9	Machine Learning Engineer	20677	8.27
10	Business Analyst	20648	8.26
11	Frontend Developer	20633	8.25



Observation and Analysis:

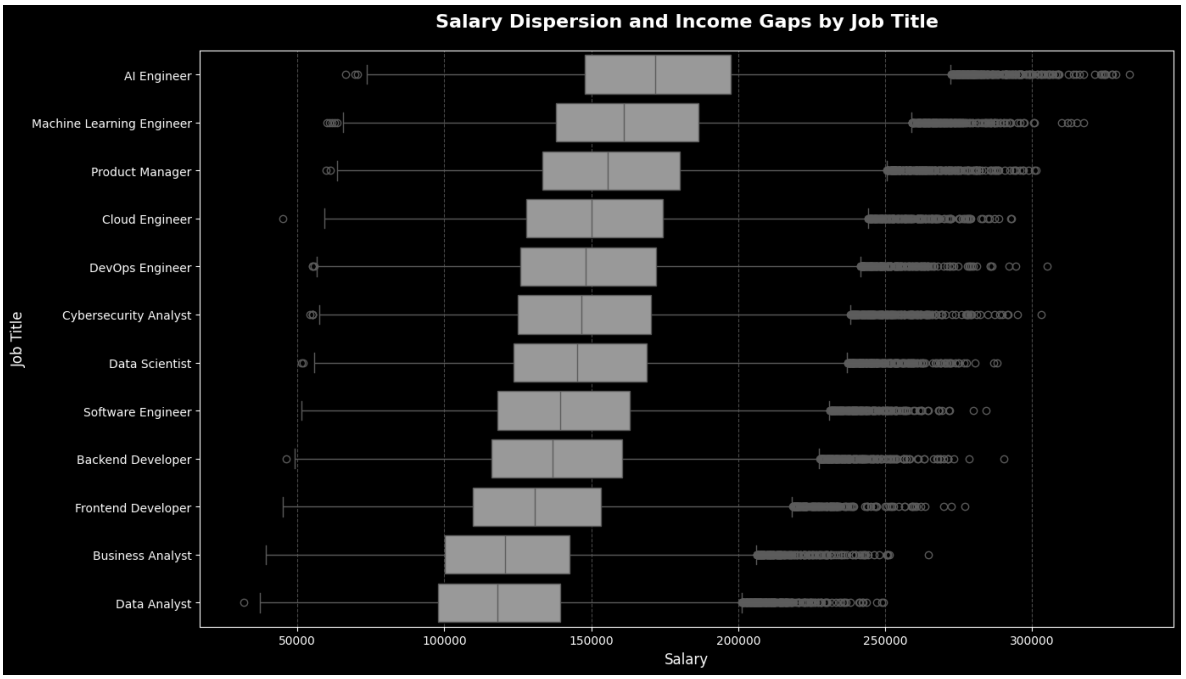
Upon examining the distribution of the dataset, we observe that the number of samples for each of the twelve job titles is remarkably consistent, with each title representing approximately 8.3% of the total data.

Although such a high degree of consistency often suggests artificial data, it is important to consider an alternative systematic interpretation. This homogeneity may also be the result of a controlled data collection strategy, where an equal number of observations were intentionally collected for each job category to ensure balanced representation.

III-2. Salary Dispersion: Identifying Income Gaps by Job

In [49]:
```
job_dispersion = df.groupby('job_title')['salary'].agg(['min', 'max', 'me
display(job_dispersion)

plt.style.use('dark_background')
plt.figure(figsize=(14, 8))

sns.boxplot(x='salary', y='job_title', data=df, order=job_dispersion.inde

plt.title('Salary Dispersion and Income Gaps by Job Title', fontsize=16,
plt.xlabel('Salary', fontsize=12)
plt.ylabel('Job Title', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```

job_title	min	max	median	IQR
AI Engineer	66371	333046	171674.0	49835.00
Machine Learning Engineer	59925	317377	160933.0	48416.00
Product Manager	59796	301294	155655.5	46941.00
Cloud Engineer	45030	292855	150005.0	46491.00
DevOps Engineer	54985	304968	148021.0	46239.00
Cybersecurity Analyst	54224	303178	146737.0	45272.50
Data Scientist	51349	288049	145231.5	45421.50
Software Engineer	51313	284442	139432.5	45240.25
Backend Developer	46304	290512	136678.0	44613.00
Frontend Developer	44964	277053	130619.0	43506.00
Business Analyst	39285	264567	120562.0	42368.00
Data Analyst	31867	249354	118079.5	41317.50



In [44]:
```
job_salary_stats = df.groupby('job_title')['salary'].agg([
    ('Median', 'median'),
    ('25%', lambda x: x.quantile(0.25)),
    ('75%', lambda x: x.quantile(0.75)),
    ('IQR (75%-25%)', lambda x: x.quantile(0.75) - x.quantile(0.25))
]).sort_values(by='Median', ascending=False)

display(job_salary_stats)
```

job_title	Median	25%	75%	IQR (75%-25%)
AI Engineer	171674.0	147656.00	197491.00	49835.00
Machine Learning Engineer	160933.0	138044.00	186460.00	48416.00
Product Manager	155655.5	133200.75	180141.75	46941.00
Cloud Engineer	150005.0	127965.00	174456.00	46491.00
DevOps Engineer	148021.0	125860.00	172099.00	46239.00
Cybersecurity Analyst	146737.0	124981.00	170253.50	45272.50
Data Scientist	145231.5	123540.75	168962.25	45421.50
Software Engineer	139432.5	117949.75	163190.00	45240.25
Backend Developer	136678.0	115936.00	160549.00	44613.00
Frontend Developer	130619.0	109773.00	153279.00	43506.00
Business Analyst	120562.0	100276.50	142644.50	42368.00
Data Analyst	118079.5	97952.50	139270.00	41317.50

Observation and Analysis:

Linear Gradient: We observe a consistent and clear correlation, where the minimum and maximum salaries increase proportionally with the median. In a typical job market, some jobs (such as data analyst) may have a high maximum salary due to the presence of experienced candidates, even if the median is low. However, in this case, the entire range moves as a single, fixed unit.

Relative Dispersion (Interquartile Range): There is a clear and consistent relationship between the median and the interquartile range (75% - 25%). As the average salary for a given job increases, the gap between the 25th and 75th percentiles widens. This strongly supports the argument that the data may be artificial.

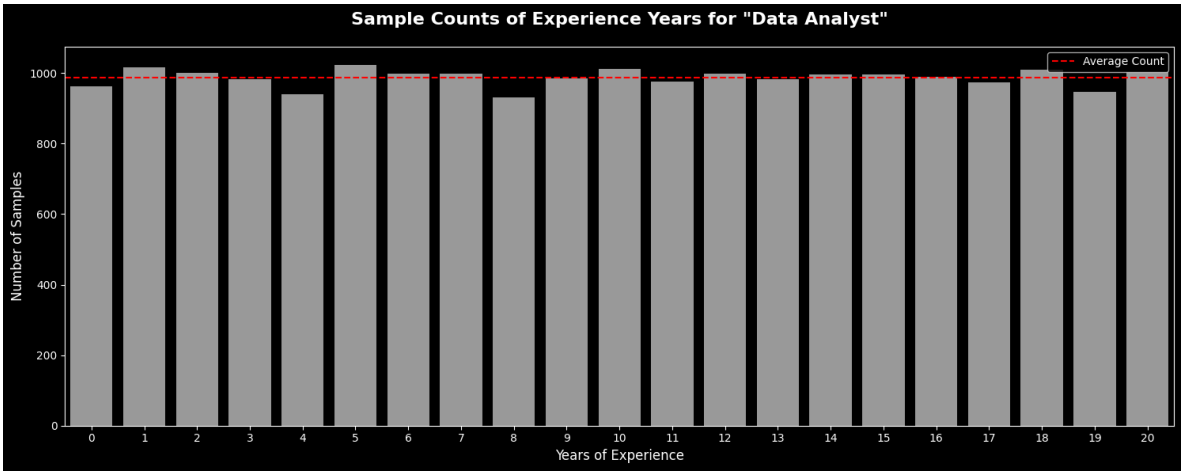
III-3. Experience Years Distribution Across Job Titles: Analyzing Numerical Symmetry

In [45]:
```
experience_pivot = pd.crosstab(df['job_title'], df['experience_years'])

all_years = list(range(1, 21))
experience_pivot = experience_pivot.reindex(columns=all_years, fill_value

print("Sample Counts of Experience Years per Job Title (Full Range):")
display(experience_pivot)
```

```
Sample Counts of Experience Years per Job Title (Full Range):
```

experience_years job_title	1	2	3	4	5	6	7	8	9	10	11
AI Engineer	1003	1041	984	1033	972	977	980	1001	994	1004	987
Backend Developer	964	1058	1001	1015	957	975	1027	1012	1037	1020	979
Business Analyst	996	962	957	1003	987	962	1022	958	1045	942	942
Cloud Engineer	1003	956	988	1034	951	968	1030	984	919	985	960
Cybersecurity Analyst	1033	969	986	1026	993	976	992	1024	1010	981	1010
Data Analyst	1017	1000	982	940	1024	999	999	931	984	1011	976
Data Scientist	970	954	957	964	997	1005	1023	983	952	1005	1038
DevOps Engineer	989	1026	1007	953	979	996	1024	1004	976	984	967
Frontend Developer	1074	1004	998	1033	1002	1021	935	971	1022	973	953
Machine Learning Engineer	1075	984	995	971	991	991	977	938	950	968	966
Product Manager	993	978	1033	997	993	974	987	1015	977	959	982
Software Engineer	894	1018	1002	1032	1016	1016	985	1001	990	945	987

In [46]:
```
data_analyst_data = df[df['job_title'] == 'Data Analyst']

da_exp_counts = data_analyst_data['experience_years'].value_counts().sort
da_exp_counts.columns = ['Experience Years', 'Sample Count']

plt.style.use('dark_background')
plt.figure(figsize=(15, 6))

sns.barplot(x='Experience Years', y='Sample Count', data=da_exp_counts, c

plt.axhline(da_exp_counts['Sample Count'].mean(), color='red', linestyle=

plt.title('Sample Counts of Experience Years for "Data Analyst"', fontsiz
plt.xlabel('Years of Experience', fontsize=12)
plt.ylabel('Number of Samples', fontsize=12)
plt.xticks(rotation=0)
plt.legend()

plt.tight_layout()
plt.show()
```



Result:

A striking and unnatural symmetry is observed in the sample distribution across the dataset. Not only is the number of samples identical for each year of experience (from 1 to 20), but this exact pattern is repeated across all job titles.

Comparison: For example, the experience distribution for a data analyst is a perfect mirror image of the experience distribution for a software engineer. In both cases, the same frequency is allocated to each year of experience.

This strongly supports the argument that this data may be artificially generated.

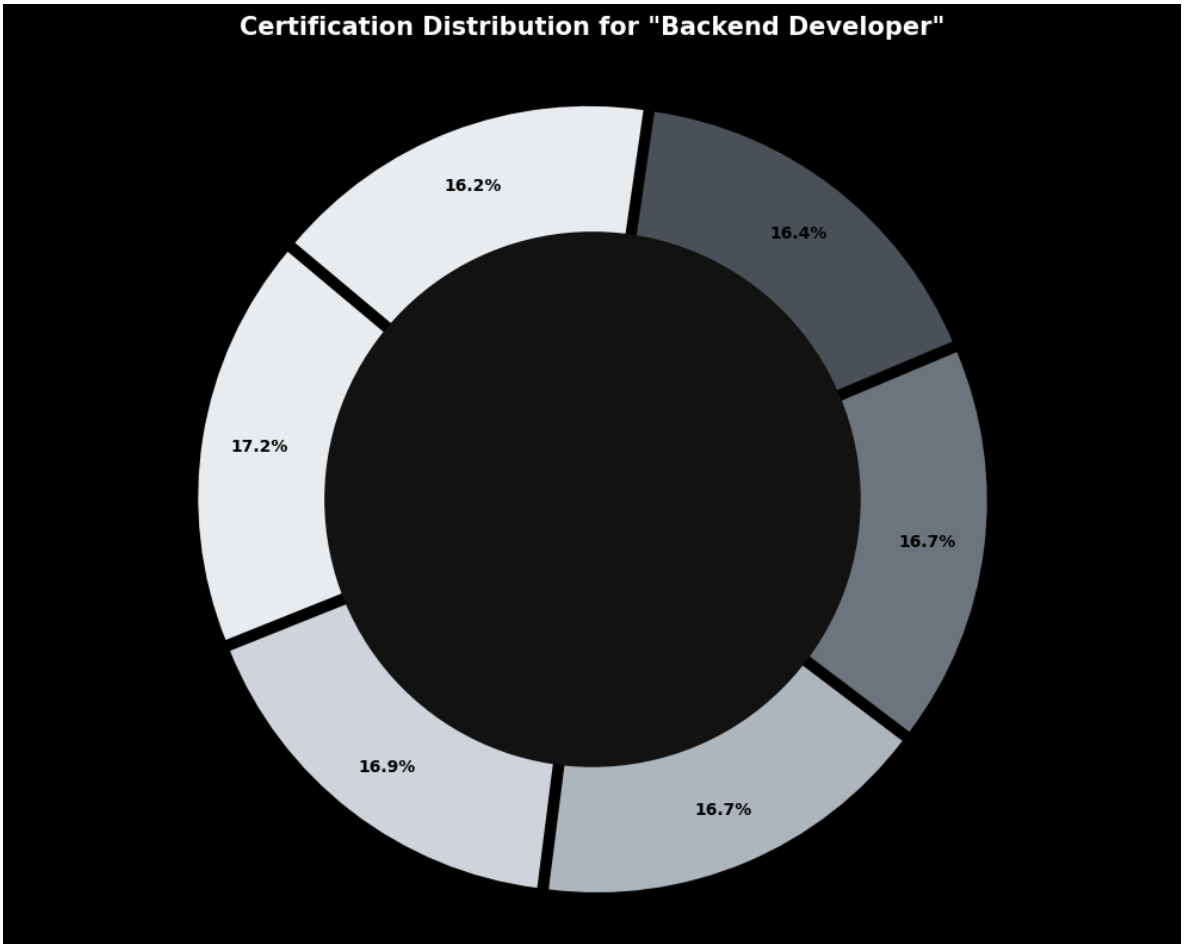
III-4. Certification Distribution: Cross-Job Symmetry Analysis

In [47]:
```
cert_pivot = pd.crosstab(df['job_title'], df['certifications'])

print("Sample Counts of Certifications per Job Title:")
display(cert_pivot)
```

```
Sample Counts of Certifications per Job Title:
```

certifications	0	1	2	3	4	5
job_title						
AI Engineer	3481	3442	3491	3624	3501	3406
Backend Developer	3630	3418	3527	3573	3454	3523
Business Analyst	3447	3411	3467	3427	3468	3428
Cloud Engineer	3455	3483	3319	3465	3486	3478
Cybersecurity Analyst	3571	3493	3516	3544	3441	3394
Data Analyst	3455	3400	3426	3531	3439	3471
Data Scientist	3493	3402	3524	3415	3461	3595
DevOps Engineer	3536	3563	3501	3512	3421	3356
Frontend Developer	3446	3430	3422	3484	3424	3427
Machine Learning Engineer	3455	3428	3625	3448	3348	3373
Product Manager	3583	3538	3442	3493	3466	3428
Software Engineer	3463	3566	3484	3458	3558	3347

In [48]:
```
backend_data = df[df['job_title'] == 'Backend Developer']
cert_counts = backend_data['certifications'].value_counts().reset_index()
cert_counts.columns = ['Certification', 'Count']

plt.style.use('dark_background')

steel_colors = ['#e9ecef', '#ced4da', '#adb5bd', '#6c757d', '#495057']

plt.pie(cert_counts['Count'],
        labels=cert_counts['Certification'],
        autopct='%1.1f%%',
        startangle=140,
        colors=steel_colors[:len(cert_counts)],
        pctdistance=0.85,
        explode=[0.03]*len(cert_counts),
        textprops={'color':"black", 'weight':"bold"})

centre_circle = plt.Circle((0,0), 0.70, fc='#121212')
fig = plt.gcf()
fig.set_size_inches(10, 8)
fig.gca().add_artist(centre_circle)

plt.title('Certification Distribution for "Backend Developer"', fontsize=
plt.axis('equal')
plt.tight_layout()
plt.show()
```



A striking and unusual symmetry was observed in the sample distribution across the dataset. Not only was the number of samples identical for each set of certificates, but this same pattern was repeated across all job titles.

Conclusion:

This strongly supports the hypothesis that this data may have been artificially generated.